LONG-TERM IMPACT OF SELECTIVE LOGGING ON TROPICAL TREE SPECIES DIVERSITY AND RECOVERY

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Abstract

Conservation of biological diversity continues to gain global relevance as a measure of sustainable forest management and a fundamental prerequisite in addressing climate change, considering the persistent threat from anthropogenic activities to forest ecosystems. This study aimed to assess tree species diversity, recovery dynamics, spatial composition and influence of soil variables following long-term selective logging in a tropical forest, compared to a primary (unlogged) forest. The study was conducted in the Budongo Forest Reserve in Uganda, where field inventory was carried out in systematically established permanent plots. A total of 204 circular plots, each covering 1,000 m^2 were spaced out at 100m intervals along a 2.5 km transect. Within each plot, four nested subplots of 4 m^2 were set up for phenology monitoring. Tree species richness and diversity between the logged and unlogged forest were evaluated using diversity indices, while species composition and similarity were assessed using non-metric multidimensional scaling (NMDS). Forest recovery was measured using the importance value index, aboveground biomass, ecological light guild, basal area and canopy cover. Additionally, the soil variables were fitted to the NMDS ordination to examine their influence on the spatial distribution of species and their composition. A total of 154 unique species were identified in the canopy layer, with a total abundance of 10872 individuals. Of these, 46 species were exclusive to logged forest, while 26 were found only in the unlogged forest. In the regeneration layer, 87 unique species representing 13047 individuals were recorded, with 22 species exclusive to the logged forest and 15 to the unlogged. This indicates higher gamma diversity in the logged forest. The differences in the diversity indices at both the canopy and regeneration layers showed higher evenness and species diversity in the logged forest. This was attributed to the coexistence of early and late successional species. The forest was found to have largely recovered, as differences in recovery variables were not significant. There was a high similarity in species composition between the unlogged forest and the regeneration layer of the logged forest, with distinctiveness only in the canopy layer of the logged forest. This was attributed to landscape variability and microclimate heterogeneity after selective logging, creating a mosaic of forest patches with unevenly distributed species from various ecological guilds, indicating a shift in species composition. The findings also revealed that soil primary nutrients such as potassium and pH are not exclusive in driving species distribution; Calcium, Magnesium, Aluminium, and Sodium also played significant roles. The study suggests selective logging can be beneficial in realizing multiple objectives of tropical forests only if carefully undertaken. While post-logging management is essential, the rotation period is equally crucial. These findings enhance our understanding of forest preservation, recovery, and restoration of perturbed and vulnerable ecosystems.

Abstrakt

Angesichts der anhaltenden Bedrohung der Waldökosysteme durch anthropogene Aktivitäten gewinnt die Erhaltung der biologischen Vielfalt als Maßstab für eine nachhaltige Waldbewirtschaftung und als Grundvoraussetzung für die Bewältigung des Klimawandels immer mehr an globaler Bedeutung. Ziel dieser Studie war es, die Baumartenvielfalt, die Erholungsdynamik, die räumliche Zusammensetzung und den Einfluss von Bodenvariablen nach langfristiger selektiver Abholzung in einem tropischen Wald im Vergleich zu einem primären (nicht geloggt) Wald zu untersuchen. Die Studie wurde im Budongo-Waldreservat in Uganda durchgeführt, wo die Feldinventur in systematisch angelegten Dauerbeobachtungsflächen vorgenommen wurde. Insgesamt 204 kreisförmige Parzellen mit einer Fläche von jeweils 1.000 m2 wurden in Abständen von 100 m entlang eines 2,5 km langen Transekts angelegt. Innerhalb jeder Parzelle wurden vier verschachtelte Unterparzellen von 4 m2 für die Überwachung der Phänologie eingerichtet. Der Baumartenreichtum und die Artenvielfalt zwischen dem geloggt und dem nicht geloggt Wald wurden mit Hilfe von Diversitätsindizes bewertet, während die Artenzusammensetzung und die Ähnlichkeit mit Hilfe der nicht-metrischen multidimensionalen Skalierung (NMDS) beurteilt wurden. Die Erholung des Waldes wurde anhand des Wichtigkeitsindexes, der oberirdischen Biomasse, der ökologischen Lichtgilde, der Grundfläche und der Überschirmung gemessen. Zusätzlich wurden die Bodenvariablen an die NMDS-Ordination angepasst, um ihren Einfluss auf die räumliche Verteilung der Arten und ihre Zusammensetzung zu untersuchen. Insgesamt wurden 154 einzigartige Arten in der Kronenschicht mit einer Gesamtzahl von 10872 Individuen identifiziert. Davon kamen 46 Arten ausschließlich in den geloggt Wäldern vor, während 26 Arten nur in den nicht geloggt Wäldern gefunden wurden. In der Verjüngungsschicht wurden 87 einzigartige Arten mit 13047 Individuen festgestellt, von denen 22 Arten ausschließlich im geloggt und 15 im nicht geloggt Wald vorkamen. Dies deutet auf eine höhere Gamma-Diversität im geloggt Wald hin. Die Unterschiede bei den Diversitätsindizes sowohl in der Baumkronen- als auch in der Verjüngungsschicht zeigten eine höhere Ebenheit und Artenvielfalt im geloggt Wald. Dies wurde auf die Koexistenz von Arten der frühen und späten Sukzession zurückgeführt. Es wurde festgestellt, dass sich der Wald weitgehend erholt hat, da die Unterschiede in den Erholungsvariablen nicht signifikant waren. Die Artenzusammensetzung des nicht geloggt Waldes und der Verjüngungsschicht des geloggt Waldes war sehr ähnlich, nur in der Kronenschicht des geloggt Waldes gab es Unterschiede. Dies wurde auf die landschaftliche Variabilität und die mikroklimatische Heterogenität nach dem selektiven Holzeinschlag zurückgeführt, die ein Mosaik von Waldflächen mit ungleichmäßig verteilten Arten verschiedener ökologischer Gilden entstehen ließen, was auf eine Veränderung der Artenzusammensetzung hindeutet. Die Ergebnisse zeigten auch, dass nicht nur Primärnährstoffe wie Kalium und pH-Wert für die Verteilung der Arten

verantwortlich sind, sondern auch Kalzium, Magnesium, Aluminium und Natrium eine wichtige Rolle spielen. Die Studie legt nahe, dass selektiver Holzeinschlag nur dann zur Verwirklichung der vielfältigen Ziele von Tropenwäldern beitragen kann, wenn er sorgfältig durchgeführt wird. Während die Bewirtschaftung nach dem Holzeinschlag von wesentlicher Bedeutung ist, ist die Umtriebszeit ebenso entscheidend. Diese Ergebnisse verbessern unser Verständnis für den Erhalt, die Erholung und die Wiederherstellung von gestörten und gefährdeten Ökosystemen.

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CHAPTER ONE

1.0 Introduction

1.1 Background to the study

Conservation of biological diversity continues to gain global relevance as a measure of sustainable forest management and a fundamental prerequisite in addressing climate change, considering the persistent threat from anthropogenic activities to forest ecosystems (Gauthier et al., 2015). Tropical forests remain the terrestrial ecosystem with high biological diversity and they play an exceptional role in regulating global climate (Hall et al., 2003). Addressing the trade-offs concerning forest biodiversity and production is therefore crucial (Asbeck et al., 2021). The timber industry, especially in the tropics adopted selective logging as a management practice to concurrently ensure production and preservation of diversity (Montoro Girona et al., 2019). Selective logging is the practice of selectively extracting valuable timber resources from the forest which are predetermined based on either diameter size classes or conservational considerations. It is the predominant harvesting practice in tropical natural forests, regarded as a single tree or group selection harvesting operation depending on the number of trees per unit area removed. This system dates back to the French and British colonial systems in West Africa, Burma, India and Malaysia (Hari Poudyal et al., 2018; Puettmann et al., 2015) and is the most widely adopted harvesting approach in most of tropical Africa, Asia and America (Martin et al., 2015).

It involves carefully controlled and guiding harvesting documents in most situations, aimed at promoting heterogeneity in microclimate, regeneration and maintenance of forest structure (Huang et al., 2020). A critical component is the intensity of this practice (number of trees harvested per hectare) which influences the recovery, species composition and the structure of the residual forest. For example, it was proposed by Bonnell et al., 2011 that, estimates of over 90 years are expected to achieve recovery, and even longer periods depending on the level of logging intensity for basal area, with species composition requiring even significantly longer periods. Thus, they observed higher diversity of tree species in moderately logged than in compartments subjected to heavy logging with regeneration of early successional species being suppressed (an indication of shift in composition). The intensity can significantly reduce biomass and stem density, promoting the spread of weeds and prolonging ecological succession for decades. At the same time, while one might expect large gap sizes to promote regeneration, they often result instead in a shift in species composition, a phenomenon referred to as 'arrested

succession' (Paul et al., 2004). Reduction in seedling recruitment and high mortality is observed due to the invasion and dominance of light-demanding species, weeds and vines in the opened canopies. In this repute, Gourlet-Fleury et al. (2013) recommend the implementation of selective logging in the direction of thinning operations for the removal of limited or few maximum DBH class species while spreading the commercially available trees to be harvested on multiple cutting cycles, to ensure the replenishment of next diameter class.

The success of selective logging as a tool to regulate disturbance intensity from harvesting is highly dependent on available measures to control and enforce its implementation. This is reflected in the recruitment or regeneration intensity of residual species in colonising the gap after removals (Win et al., 2012). Considering the implementation of selective logging in the tropics and the management demands in harmonizing timber production, conservation of biodiversity and carbon sequestration, efficient activity management is imperative to realize these expectations. The minimum harvestable diameter is a key criterion adopted under reduced-impact logging (RIL) practices to ensure continuous forest cover and minimize resource depletion from logging. The concept of RIL as used in most contexts of polycyclic management systems in the tropics arose as an initiative to address the forest degradation dilemma arising from the inefficiencies of conventional selective logging in adhering to the recommended practices. Therefore, reduced-impact logging can be regarded as a form of selective logging at different intensities and is considered in other literatures as an improved implementation of forest harvesting (Putz et al., 2008). For example, assessing the disparity of species composition in conventional logging, reduced-impact logging (selective logging) and primary forest revealed that, for a well-regulated activity, it is efficient as a sustainable harvesting tool (Imai et al., 2019). The adoption of selective logging (RIL) with its accompanied regulations aims to promote the coexistence of late and early successional species to elevate the diversity of the forest. This is well explained by Imai et al., 2012, that on a larger spatial scale, species richness was higher in RIL than in the primary unlogged forest, corroborating also with the outcomes from Menge et al., 2023 that beta diversity is higher in managed forests compared to unmanaged temperate forest. Despite the operational concerns and complications, selective logging is regarded as a sustainable forest management approach in tropical natural forest management to ensure production (resource extraction) while considering intensively the integrity of the forest or the recovery ability of the residual forest. The acceptance of selective logging under the Forest Stewardship Council (FSC) as a sustainable forest management practice and adoption in reducing emissions from deforestation and forest degradation (REDD+) programs in Africa speaks a lot about its reputation, however

elucidating the long-term impact associated with forest recovery remains inadequately explored (Cazzolla Gatti et al., 2015).

Understanding how the establishment and distribution of tree species are influenced by the alteration of soil properties from logging is a crucial component in assessing forest recovery after logging. Soil nutrients and properties significantly influence tree growth, distribution and compositional shift in species in the forest. The soil component considered is expected to complement the pattern illustrated by the variability in diversity. Understanding how these soil properties are driving the dominance and phasing out of certain tree species, and determining the compositional shift after logging is therefore imperative. The critical role of soil primary nutrients (Nitrogen (N), Potassium (k) and Phosphorus (P)) in this process cannot be overemphasised and several studies have highlighted this. It is therefore relevant to concede that soil NPK is vital in influencing the distribution of species in the tropics, however, it is also prudent to appreciate that these studies revealed potassium(K) as the predominant factor in this reputation(Umaña et al., 2021). For example, (John et al., 2007) revealed that K has the strongest effect on community structures. In addition to this, Manu et al., (2022) found that K and N are the essential nutrients influencing the productivity in the Budongo forest reserve. Notwithstanding, Condit et al., 2013 and John et al., (2007)presented the absence of a strong response of species to nitrogen in the tropics attributed to its availability and further accumulation through atmospheric deposition. Several studies have also unveiled the imperative role of soil pH in this discourse (Hong et al., 2018; Liu et al., 2021). This is also acknowledged by Okimat et al., (2023) and addresses the relevance of considering other soil variables to complement the role of pH in the study area. Towards this direction, pH and K are considered proxies for evaluating the extent to which Aluminium (Al), Calcium (Ca), Magnesium (Mg), Sodium (Na), Cation Exchange Capacity (CEC) and Iron (Fe) drive forest productivity. These are essential but less considered soil variables in forest research (USDA - NRCS, 2019). CEC is the measure of total cation held by the soil due to the presence of anionic exchange sites and released into the soil solution to be available for plant use (the ability to hold and exchange cations) (Saidian et al., 2016). The compositional shift in species distribution and therefore spatial variability in species coexistence is expected to be influenced by the availability and condition of soil variables such as Al, CEC, Fe, Mg, K, Na, Ca and soil pH.

1.2 Problem statement

Assessing tropical tree species diversity under selective logging is crucial because it is the fundamental determinant of forest structure and processes, and the diversity of other taxonomy is directly determined by the presence or absence of these tree species as resources, and the extent to which they provide and influence habitats. The debate surrounding the impact of selective logging on ecosystem multifunctionality remains unresolved. Like other forms of forest disturbances or removal, selective logging alters canopy structure, biomass and species composition, with the extent of these changes largely influenced by the intensity of logging (Okuda et al., 2003; Rutishauser et al., 2016). While many studies report that an increase in logging intensity is detrimental to the function and recovery of the forest, other studies suggest otherwise (Gibson et al., 2011; Huang et al., 2020; Putz et al., 2012). For example, Huang et al., 2020, on assessing ecosystem multifunctionality from selectively logged forests subjected to different intensities reported that the intensity of selective logging was positively correlated with several ecosystem functions (i.e. nutrient cycling, soil carbon, decomposition and wood production). This study suggests the increase in functional diversity with higher intensities of selective logging, which is unsurprising, given that the study area is dominated by a fast-growing, light-demanding timber species. However, biological diversity, which was less considered, showed a negative correlation. This has been the trend in most studies associated with the impact of selective logging on species diversity and other ecosystem services (inadequate report on other ecosystem functions). This is one of the challenging areas related to the performance of selective logging as a practice to address the concept of sustainable management. To ensure sustainable production of timber products, the extent to which the commercially valuable species are managed under this harvesting operation and the measures to ensure their availability and establishment after logging is crucial.

Another challenge in research efforts to quantify the effects of selective logging on biodiversity is the extended period required for tropical forests to fully recover after logging. As a result, many studies conclude that the post-logging period is insufficient for a complete species diversity recovery assessment in tropical forests. For example, Bonnell et al. (2011), Gaui et al. (2019), Gourlet-Fleury et al. (2013), Shima et al. (2018), and Do et al. (2016) conducted recovery studies 17, 25, 24, 40, and 30 years after logging, respectively. The success of conservation measures and management of harvesting cycles in tropical forests revolves around the level of certainty of forest recovery and the period required. A knowledge gap exists on the expected number of years for recovery after selectively logging a tropical forest with regard to a defined level of intensity.

Soil is a crucial component underpinning the process of germination, establishment and distribution of tree species in tropical forests. The obvious evidence for soil properties' effect on tree species' spatial distribution is illustrated in studies conducted at larger landscape scales. The distinction in the heterogeneity and clear segregation in edaphic types as a result of distinguishable mosaics of soil properties at this scale of assessment makes it relatively undemanding in quantifying the effect of soil properties on the spatial distribution of tree species (Phillips et al., 2003; Tuomisto et al., 2002). This is challenging and less clearly presented in literature when the study is at a small local scale, especially plot level due to spatial aggregation. The conspicuousness of this redistribution is not clearly illustrated in the literature on tropical forests, especially at a small local scale (plot level) in sampling studies. In a highly diverse tropical African forest ecosystem like the Budongo reserve, tree growth and establishment are dictated by multiple soil nutrients and their combinations. Heightened emphasis has been on considering the primary nutrients and soil pH (Manu et al., 2022; Okimat et al., 2023)with minimal exertion channelled into exploring how critical other variables like Al, Fe, Ca, Mg, Na and CEC can be relative to the primary nutrients and soil pH. The influence of soil primary nutrients and pH are crucial in dictating the spatial composition of species in the forest, however, is this an exclusive role or do other soil variables have the tendency of driving comparatively similar effects? After subjecting a portion of the reserve to selective logging, there have not been long-term assessments of the impact of logging on regeneration dynamics, species composition and recovery of the residual stand. In the absence and knowledge of successful restoration, these sites become exceptionally vulnerable to conversion to other land uses recognized as more profitable. The tropical high forest of Uganda has declined drastically with an even greater threat to the highly valuable timber species. Understanding the dynamics revolving around tropical forest recovery is paramount in conservation measures and decisions on the economic values of the forest. Intervention can be successfully implemented when adequate knowledge of the dynamics of recovery and species distribution after selective logging is available. This study strives to bridge the knowledge gap in this regard. This is also essential in informing restorationists of the expected recovery status of tropical forests at a certain point in time and the impact on the species involved after selectively logging the forest and further elevating understanding of species regeneration ecology. This is crucial in elevating our knowledge in preserving, recovering and restoring perturbed and vulnerable forest ecosystems.

1.3 Aim and Objectives

1.3.1 Aim

This study aimed to assess the long-term impact of selective logging on tree species diversity and recovery in the Budongo Central Forest Reserve, Uganda.

1.3.2 Specific Objectives and Hypotheses

The objectives and hypothesis are as follows:

1. Assess the tree species composition, diversity and relative recovery dynamics after a long period of logging
    i. Species diversity is higher in unlogged forests than in logged forests even after a long-term recovery.
    ii. The forest biomass and the dominating canopy trees recover to some extent after long-term disturbance similar to the unlogged forest.
2. Determine the similarity and dissimilarity in species composition of logged and unlogged forests.
    i. Do overstory trees regenerate well or does a different regeneration composition suggest a long-term species compositional shift?
3. Assess the influence of soil properties on the spatial distribution and composition of tree species
    i. Soil pH and Potassium (K) are the main driving factors of the spatial distribution of species in the logged and unlogged forest.

CHAPTER TWO
2.0 Literature review

2.1 Forest Management Systems

2.1.1 Introduction to Forest Management Systems

The forest management systems have gone through a series of changes over the years with some management systems completely disregarded in natural forest management. Systems associated with concepts such as close-to-nature forestry, retention forestry, continuous cover forestry and ecological forestry are promoted in the management of natural forests. This disregards even-aged forest management or monocyclic systems like clear-cutting and shelterwood systems. The management system predominant in tropical natural forestry is the selective logging system which includes group and single tree selection. The contrasting extreme is where the forest is passively managed through the allocation of land areas as strictly protected. Protected forests are sometimes referred to as unlogged or primary forests (no human intervention) depending on past managements associated with them and they continue to be managed uniquely to ensure their reputation, including boundary management and control of invasive species. In the tropical setting, conservation is best achieved with outright protection by directly enacting strict policies to ensure its implementation (Nasi & Frost, 2009). The interesting thing is that, in many tropical countries, what was once deemed a protected forest and highly biodiverse has already lost its value due to illegal encroachment with some already lost (Laurance et al., 2012; Potapov et al., 2017). This then makes it crucial to reassess our perception of the existing forests regarded as unlogged or protected in their biodiversity conservation status in certain periods, a precautionary measure toward certainty. Another prevailing management system already introduced is natural forest management under a selective system. The objective is mostly sustainable timber production depending on the intensity of removal while ensuring the structure and diversity of the forest. Depending on the management plan, species based on accepted diameter class are removed within a specific cutting cycle and the forest is allowed to recover. The shelter or shelterwood system is also deployed in natural forest management in the tropics and used in other literature to depict selection systems. Unlike the selection system where the creaming of desirable commercial species is implemented, more relevance and effort in retaining parent seed-bearing trees and trees with larger crowns to supply shelter for the next cohort of species is highly regarded under the shelter system. The system deploys a series of silvicultural activities that enable conditions for the germination, growth and establishment of species in different guilds. This is essential in natural forest management, especially for medium light requirers or gap opportunists who are in most situations disadvantaged in less intensity or high-intensity logging. Tree species

diversity is therefore enhanced when species from all categories are fairly represented in the landscape (Ashton et al.,2011;Putz, 2011). Additionally, emphasizing one system as optimal over the other based on sustainability in the case of shelterwood and selection system can be trickier.

In as much as an argument can be made for shelterwood for timber yield, cost efficiency, vitality and diversity of remaining forest after the activity, a scrupulously undertaken selective logging can guarantee nothing less of that. While it is not debatable the effect that conventional selective logging subjects the natural tropical forest to, a new dimension rewriting the perception of this system is the reduced impact logging (RIL) which strives to reshape the silvicultural activities under the selection system and implement techniques to minimize ecological damages and to improve resilience such that the remaining forest regain that earliest capacity to recover (Sist et al., 2003). This can be regarded as a recent remedy to address what was once solely meant for the removal of all commercially valuable timber species from the forest. It has been conspicuous over the years how modern silviculture for example has evolved to a more sophisticated adaptive activity incorporating ecosystem management with relevance to sustainability to what was once a simple exploitation practice. Therefore, incorporating the selection system under the RIL in conservation activities is feasible when the regulations and directions governing the activity are meticulously respected. The Shelterwood and selection system (with RIL) executed within an acceptable or ethical intensity is therefore suitable for ensuring the retention of a large proportion of biodiversity and conservation of natural tropical forests (Günter et al., 2012). While this concept will be thoroughly explained under selective logging, the main point to highlight here is that is a management system for natural forests (Putz & Thompson, 2020). The success of the management systems is driven by a spectrum of interventions referred to as silvicultural activities implemented before, during or after logging. The silvicultural activities are embedded in the management systems and therefore cannot be excluded as a component. In the management of natural forests, these include but are not limited to; pre- and post-harvesting activities like cleaning and liberation cuttings, controlled or assisted natural regeneration, improvement and commercial thinning and enrichment plantings (Günter et al., 2012). Silviculture according to ITTO 2002 is defined as: “the art and science of producing and tending forests by manipulating their establishment, species composition, structure, and dynamics to fulfil given management objectives”.

2.1.2 Sustainable forest management and its importance

The manager's objectives and the existing policies widely drive the deployment or adoption of a particular forest management system. Over the years, there has been a modification in the management systems and to some extent discarded to adapt to changing conditions, especially to the climate. To establish a particular line of action and a direction regarded as optimal for all groups of people, a standard recommendation of forest management was then developed with its criteria and indicators to serve as a measuring and reference tool. This is termed sustainable forest management. Several activities are adopted under the term to operationalise the concept due to that, all the definitions from different agencies and forest stewards have a common expected outcome. The Ministerial Conference on the Protection of Forests in Europe (MCPFE) defined sustainable forest management as: “The stewardship and use of forests and forest lands in a way, and at a rate, that maintains their biodiversity, productivity, regeneration capacity, vitality and their potential to fulfil, now and in the future, relevant ecological, economic and social functions, at local, national, and global levels, and that does not cause damage to other ecosystems.”(MCPFE, 1993). The Montreal Process criteria also defined sustainable forest management as "Maintenance of forest ecosystems health and vitality, of the productive capacity of forest ecosystems, of forest contribution to global carbon cycles and of long-term multiple socioeconomic benefits to meet the needs of society, conservation of biological diversity and soil and water resources" (Montreal Process, 2009). From a different organisational perspective, The International Tropical Timber Organisation defined sustainable forest management as: “the process of managing forests to achieve one or more specified objectives of management concerning the production of a continuous flow of desired forest products and services without excessive reduction of its inherent values and future productivity and undue undesirable effects on the physical and social environment.” (ITTO, 2014). The key and common component obvious in the definitions is the emphasis on a good management procedure that ensures the continuous flow of forest services and functions without compromising the vitality of the forest, which largely is dictated by the efforts to ensure biodiversity (Köthke, 2014). ITTO has been an integral organisation in the tropics, ensuring ethical management processes in all aspects of tropical natural forests, especially in trade and resource usage, conservation and sustainable management. The sustainable management of tropical forests revolves around the criteria and indicators developed by the ITTO policy development series which serves as a tool for assessing and monitoring the conditions of tropical natural forests. From the assessment of the recent criteria and indicators published in 2016, five of the seven criteria are meant to address biodiversity in natural tropical forests

(ITTO, 2016). Biodiversity in natural forests cannot be achieved with the neglect of tree species diversity because it is the fundamental determinant of forest structure and processes, and the diversity of other taxonomy is directly determined by the presence or absence of these tree species as resources, and the extent to which they provide and influence habitats. This narrows down the definition of SFM to the relevant component of resources in the forest and ensures the implementation at all stages in tree development to maximize the expected services and functions of the forest. This involves concurrently being critical of tree species diversity, acknowledging and meticulously managing species of commercial value, elevating regenerative capacity, strictly protecting distinct species and ensuring the perpetuity of these functions and services. The main pillars of SFM are ecological, socioeconomic and technological aspects. The term has transformed the direction of the dominant aim of forest perceived as a source of just timber and non-woody forest products in the past to an inclusive service where the environment is crucial in recent years. With timber production being the sole objective of the forest, this implies that the silvicultural systems developed many decades ago were targeted to this course. This then questions the other pillars of sustainability and forest management in recent years where the objective of the forest is valued in other perspectives beyond timber extraction. The response is the modifications and relinquishing of activities deemed detrimental to addressing all the aspects of sustainable forest management. SFM is a mere concept unachievable in the absence of appropriate silvicultural activities incorporated under the various forest management systems. Changing the narrative therefore to link silviculture to be more societal and ecologically oriented can be effective towards this course(Günter et al., 2012).

The implementation of SFM over the years has not been a smooth journey as expected though there is the acknowledgement as the appropriate approach to forest management. Several factors account for this retrogress in the tropics including, weak policy enforcement and governance accelerating illegal activities, poor land tenure and ownership rights, overdependency on the forest, competing use of land resulting in degradation and deforestation, high cost of forest retention and conflicts. The local demand for wood products in most tropical countries exceeds even the state exploitation of timber and this has been challenging to monitor under SFM due to complications in developing a legal system to control the activities. For example, in Ghana, it is approximated that 80 percent amounting to 6 million m^3 of harvested timber is a product from chainsaw logging to meet the domestic demand for wood though this activity is declared illegal. It is alarming especially when the monitored harvesting accompanied by the required documents for the state is just 2 million m^3 (Hansen & Treue,

2008; Hawthorne et al., 2012). Similar challenges are encountered in the attempt to realise the SFM objectives in different tropical countries therefore hampering the success. Notwithstanding, recent efforts in other tropical countries to be strict with protected forests are commendable, however many of these forests were heavily exploited in the past with little consideration for the future, therefore, the diversity and productive capacity of these forests are mostly compromised leading to rapid extirpation of their commercial value. Many such species like mahoganies are now red-listed or classified as endangered under CITES (Convention on International Trade in Endangered Species of Wild Fauna and Flora) (Okimat et al., 2023). The aspect of regeneration capacity has received less attention over the years in addressing the management of natural tropical forests. Interventions to liberate and stimulate the growth, establishment and development of species after disturbances in the forest have been disregarded resulting in vast areas colonised by invasive alien or weed species. An attempt to address these impediments therefore requires a collaborative effort at all different levels. From a policy perspective, the successful introduction of forest certification in the 1990s and other legal frameworks was an intervention to ensure the marketing of legally approved forest products satisfying the standards of ecological and social development (Günter et al., 2012). The availability of information on the assessment of the current state of the forest resources, especially tree species diversity is also crucial (Rutishauser & Herold, 2020).

2.1.3 Forest management and biodiversity conservation

The complex ecosystem functions and services provided by the forest cannot be overemphasized, from a wide range of human well-being contributions to the economy, environment and global climate processes. The functional process and biological diversity of the forest ecosystem are the backbone of sustainably ensuring the stream of these services, and excellent knowledge about this is crucial. The knowledge only emanates from forest management which ensures continuous connection with the ecosystem to ascertain the evolution and response process through time involving ways to approach contemporary challenges. Forest management and biodiversity are therefore interconnected, either active management is required in situations where there is species culmination or asymptote, species invasion or promotion of certain species, or inactive management to allow nature to take charge (Bohn & Huth, 2017). Ensuring the sustainability of forest resources by harmonising

production with biological diversity through the conservation of other forest landscapes are just a few obstacles to address in this sector. This makes it more demanding in this perspective to deploy a particular forest management activity in the hope of addressing the conflict of interest of all stakeholders and ultimately conserving diversity. Forest ecosystems can be managed strategically to enhance biodiversity, especially in depleted landscapes. For species-depleted landscapes like secondary forests, there is no doubt how instrumental active forest management is required to restore biodiversity however reservations are observed in the case of primary forests. A classic example of biodiversity conservation will be the identification of strategically located primary forests with high levels of species diversity and endemism and strictly prohibiting access (Angelstam et al., 2018; Gering et al., 2003; Wang et al., 2019). In as much as this sounds interesting but very challenging to realise, another dimension to assess this is whether an untouched primary forest can be confidently regarded as the ecosystem with the highest species diversity. There is contrasting opinion in the literature on whether the species diversity of primary forests can be elevated with forest management interventions or should be left as they are. In each situation, a thorough understanding of the complexities of the forest is highly obligatory. (Gustafsson et al., 2020; Jandl et al., 2019). A study by (Asbeck et al., 2021), on measuring the biodiversity response to forest management intensity revealed an abrupt relative change in species composition or richness with an increase in management intensity. The findings from (Dinca & Zhiyanski, 2023) also illustrate that the species diversity after management abandonment showed an initial consistent development to a phase where there was less diversity and dominance of shade-tolerant species. Results from (Ravnjak et al., 2022) also corroborate the earlier findings that intervention improves species diversity with a confirmation that absent species appeared when forest gaps were created in the forest. Associating species diversity to microclimate heterogeneity, (Menge et al., 2023) discovered higher diversity at the landscape scale in the managed forest compared to unmanaged, attributing it to spatially co-existing different stages of forest development resulting from management activities altering the understorey microclimate. This is not to say disturbance at any intensity is tolerable by the forest ecosystem, but any treatment meant to be administered should be cautious and consider the conditions of the remnant forest and its ability to shock tolerance. Notwithstanding, the reputation of species diversity in tropical primary forests is exceptional and if management is recommended, should be extremely cautious.

2.3 General Overview of Selective Logging

2.3.1 Selective Logging in the Tropics: Practices and Principles

In the tropics, natural forest management has been influenced by silvicultural practices developed many years ago, primarily focusing on the selective extraction of one or a few commercially valuable species within a specified rotation or cutting cycle. Historically, the emphasis of tropical forest management has been predominantly on timber production, with the main goal being the sustainable yield of timber. These management practices were also accompanied by measures to remove or eliminate non-desirable species, a common practice detrimental to the vitality of the forest and the environment through the adoption of chemical girdling or tree poisoning. Selective logging is the practice of selectively extracting valuable timber resources from the forest which are predetermined based on either diameter size classes or conservational considerations. It is the predominant harvesting practice in the tropics, regarded as a single tree or group selection harvesting operation depending on the number of trees per unit area removed. The key strategy involved harvesting trees that exceeded a minimum diameter and allowing the forest to regenerate for a 30 to 40-year rotation cycle. Consequently, a critical question for foresters has been the rate of timber stock increase during this rotation cycle. According to this management paradigm, sustainability is achieved if the volume of timber harvested in each cycle matches the volume that will regenerate over the following 30 to 40 years.

Typically, the most valuable timber species are harvested first, followed by progressively fewer valuable ones in subsequent cycles, a practice known as 'logging down the value chain'. Moreover, most governmental and non-governmental certification bodies, such as the Forest Stewardship Council, lack the resources to ascertain whether yields from specific species or entire forests are being maintained. Under current regulations in most tropical countries, timber stocks fail to recover primary forest volumes by the end of each legally designated minimum-cutting cycle, and achieving similar conditions often requires extensive time. For instance, conventional timber harvests in Amazonian Brazil showed that timber volumes needed over 60 years to recover, contrary to the 25-30 years mandated by law for the cutting cycle (Vidal et al., 2020). Another critical aspect is the yield recovery following these harvesting activities. A meta-analysis of over 100 publications on yield recovery (Putz et al., 2012) revealed significant variability but concluded that timber yields decreased by approximately 46% from the first to the second harvest. This study also found that, on average, 76% of carbon is retained in forests logged once, and 85-100% of mammal, bird, invertebrate, and plant species persist after logging, though long-term survival is uncertain. While these findings do not absolutely indicate

the complete jeopardization of forest resources and biodiversity, they can be misleading because they pertain to primary forests that have undergone initial selective logging, thus influencing outcomes in subsequent visits. It is crucial to recognize that such studies typically report on a limited number of taxa and often overlook broader ecosystem functions, particularly those delivered by complexes of co-evolved species. Despite many tropical forests undergoing second or third harvests, most reviewed studies focused on timber extraction from primary forests. A consistent finding is that despite the conservation potential of extensive selective logging, sustainable forest management (SFM) is compromised in many areas of the tropics due to poor logging practices (Ellis et al., 2019) and premature re-entry logging of previously harvested stands (Sasaki et al., 2016). These cumulative effects deteriorate the reputation and condition of forests, particularly primary tropical forests, leading to a highly degraded landscape within a few years of timber extraction. The term 'degraded' becomes unavoidable when once-prestigious primary tropical forests are reduced to secondary forests with diminished ecosystem functions and services. Labelling selectively logged forests as degraded suggests they are less competitive with other land uses, making them more vulnerable to being converted to non-forest uses. Though reflecting on these activities, there are regrets, especially from the biodiversity perspective how the effort was channelled in eradicating certain species of relevance in recent years, notwithstanding, assessment from the timber point of view was regarded a success. It is evident that if logging and access are properly controlled, these secondary effects can be eradicated, allowing extensive areas of selectively logged forests to retain significant conservation value (Roopsind et al., 2018).

The value of extensive production forests for biodiversity and ecosystem services varies with logging intensity (França et al., 2017), logging methods (Pinard and Putz, 1996; (Vidal et al., 2016), and particularly with post-harvest secondary effects such as deforestation, poaching, and illegal logging. As already highlighted, these were less considered under the objectives of selective logging, but the narrative is modified and keeps receiving alterations. To ensure continuous cover and disregard for resource depletion from logging, the minimum harvestable diameter under the reduced impact logging (RIL) is a criterion adopted under this activity. The concept of (RIL) as used in most contexts of polycyclic management systems in the tropics arose as an initiative to address the forest degradation dilemma arising from the inefficiencies of selective logging in adhering to the recommended practices. Therefore reduced-impact logging can be regarded as a form of selective logging at different intensities and considered in other literature as an improved implementation of forest harvesting (Putz et al., 2008). Therefore, the timber industry, especially in the tropics implements selective logging as a

management practice to concurrently ensure the production and preservation of diversity ensuring meticulous implementation of the reduced-impact logging practices (Montoro Girona et al., 2019). It also involves carefully controlled and guiding harvesting documents in most situations, aimed at promoting heterogeneity in microclimate, regeneration and maintenance of forest structure (Huang et al., 2020). Since reduced-impact logging often equates to reduced income, it is unrealistic to expect loggers to adopt better harvesting practices solely out of enlightened self-interest (Putz et al., 2000). However, exploring ways to offset the opportunity costs associated with improved practices to reduce timber extraction and enhance forest health is crucial. Payments for ecosystem services (PES) appear to be a promising mechanism to promote sustainable forest management (SFM), although successful applications of this tool are limited, and the benefits tend to be short-lived and reliant on continuous funding. Nonetheless, the benefits of stringent enforcement are significant. The United Nations Reduced Emissions from Deforestation and Forest Degradation (REDD+) program supports carbon crediting from improved forest management, yet funding remains limited for transitioning from forest exploitation to sustainable management(Putz & Thompson, 2020). Despite the operational concerns and complications, selective logging under the reduced impact logging is regarded as promising towards sustainable forest management in tropical natural forest management to ensure production (resource extraction) while considering intensively the integrity of the forest or the recovery ability of the residual forest. Therefore, various interference and efforts from stakeholders who have successfully rewritten the management practice deserve to be commended though expectations have not been realised to the fullest. Recognizing selective logging under the Forest Stewardship Council (FSC) as a sustainable forest management practice and adopting REDD+ programs in Africa speaks a lot about its reputation(Cazzolla Gatti et al., 2015; Putz & Thompson, 2020).

2.3.2 Impact of Selective Logging on Species Diversity

A critical component of selective logging is the intensity of this practice (number of trees harvested per hectare) which influences the recovery, species composition and the structure of the residual forest. At each level and phase of implementation, the effect is either exerted on a commercially valuable species through the creaming process or the eradication of species regarded as nuisance. The composition of tree species is influenced at all growth stages from the seed bank, seedlings, pole and matured stage by the initial operations of road construction to timber felling, extraction and soil interference. The changes in species composition can be assessed in two dimensions, changes imposed directly on the tree species and changes arising

due to modification of conditions necessary for germination and growth reflected in the recruitment or regeneration intensity of residual species in colonising the gap after removals (Win et al., 2012). From the intensity of logging and its impact on forest recovery and species diversity, Bonnell et al., 2011 presented in their findings that estimates of over 90 years are expected to achieve recovery, and increased periods depending on the level of logging intensity for basal area, with species composition requiring even significantly longer periods. Higher diversity of tree species was observed in moderately logged than in compartments subjected to heavy logging with regeneration of early successional species being suppressed (an indication of shift in composition). The intensity can significantly decline the biomass and stem density and conducive promotion of the spread of weeds thereby prolonging the required ecological succession for decades. As it would be expected that large gap sizes should promote regeneration, in most situations, there is rather a change in composition as termed by Paul et al., 2004 as arrested succession. Reduction in seedling recruitment and high mortality is observed due to the invasion and dominance of light-demanding species, weeds and vines in the opened canopies. In this repute, Gourlet-Fleury et al., 2013 recommend the implementation of selective logging in the direction of thinning operations for the removal of limited or few maximum DBH class species while spreading the commercially available trees to be harvested on multiple cutting cycles and ensuring the replenishment of the next diameter class.

Although extensively managed forests can support a high degree of biodiversity, selective logging can still have substantial negative impacts on populations of high-value tree species and the associated fauna. This is partly due to the loss of seed sources and dispersal agents. Recovery of these populations often requires meticulously managed planting (enrichment planting), a process that is challenging to implement in natural forests. The success of germination, growth and establishment of species in the forest post-logging is highly determined by the system adopted and its silvicultural practices implemented. For example, assessing the disparity of species composition in conventional logging, reduced-impact logging (selective logging) and primary forest revealed that, for a well-regulated activity, it is efficient as a sustainable harvesting tool (Imai et al., 2019). The adoption of reduced-impact selective logging with its accompanied regulations aims to promote the coexistence of late and early successional species to elevate the diversity of the forest. This is well explained by Imai et al., 2012, that on a larger spatial scale, species richness was higher in RIL than in the primary unlogged forest, corroborating also with the outcomes from Menge et al., 2023 that beta diversity is higher in managed forests compared to unmanaged temperate forests.

2.4 Tree species diversity

The variation and variability within and between life at all interpretation levels possible are represented or expressed by biological diversity, from taxonomic diversity (species to the highest hierarchy), functional diversity (trait expression and peculiarity in niche), population and to ecological diversity in the ecosystem. This is not excluded for trees in the forest, and the diversity can be assessed at the regeneration or canopy layer, between adjacent landscapes and even at the plot level. Biological diversity exists in conjunction with biotic and abiotic conditions attributing a pattern of distinctness in species, habitat, function and composition in the ecosystem at the local (assessment plots) and landscape level. The Biological Diversity Convention officially define biodiversity as the variability between and within living organisms from all sources from aquatic ecosystems to terrestrial and the complexities of the ecological components of which they are involved. Over the years, indicators of biodiversity are expressed from species diversity indices which communicate the diversity of an ecosystem mathematically from scientifically proven procedures. The well-known diversity indices present the concept of a species and ecological diversity from species richness to the proportionate distribution of species in samples and ecosystems (Fedor and Zvaríková 2019).

2.4.1 Measure of species diversity (Diversity indices)

The diversity indices are approaches that thrive on quantifying the diversity of the study area for easy interpretation and presentation. For a comparative study, glancing at these values fairly explains the individual study sites or the plots under consideration. The diversity indices revolve around two main concepts or diversity components: species richness and species evenness. This denotes that, a value for a diversity index is an attempt to explicate how species are rich or even within the study plots for the ecological study. Diversity studies involving quantitative expression of diversity mostly commence at the species-level assessment of alpha, beta and gamma diversity. **Alpha diversity** is also referred to as species richness at the local or plot level in other literature and it simply involves the counting of unique or different species present in the area, habitat or sample plot, denoted by the letter α. **Beta diversity** explains or measures the diversity between the different plots or habitats. It can then be rephrased that, beta diversity compares the alpha diversity of two plots, habitats or ecosystems within a particular landscape, and the symbol denotes it β. Within this context of an ecological study where numerous plots were equipped, the traditional computation of β diversity is discarded and the dissimilarity matrix from the Jaccard index or Bray Curtis is presented. Projecting the assessment of species from the plot or habitat level to the landscape scale is then referred to as

gamma diversity. **Gamma diversity** is achieved as a result of a linear combination of alpha and beta diversity (Alpha + Beta). Therefore, this is the total species richness in the ecosystem or landscape under study and can be said to be the alpha diversity of the entire study area. The symbol γ denotes gamma diversity (Shil and Thakare, 2023).

For the ecological indices, an extensive variety exists that endeavours to represent numerically the species richness or evenness within and between plots of study, and as it can seem challenging to select or decide the best model to measure biodiversity, there is absolutely no universally suitable or preferable index of biodiversity in ecological research. The choice of a particular diversity index mostly stems from how straightforward it can be interpreted without requiring a piece of special knowledge to understand the values generated, and this is the motivation behind certain indices widely adopted in the literature. In a sampling study, single-species richness is an elementary approach to express species diversity demonstrating the total number of species in the sample, denoted with the symbol S. This is a very useful and informative output in basic ecological studies to know how many different species are represented in the sample plot or landscape, however, it appears less informative and quite insufficient in its interpretation beyond the values provided. This therefore necessitates ecological studies to express species richness beyond this by adopting sophisticated form of species richness indices. The species diversity indices widely deployed in ecological studies emphasise on species distribution within the area under study articulated as relative abundance or dominance. Based on good or moderate discrimination, fair sensitivity to sample size, emphasis on richness or evenness and also involving a simple understanding of the output, the most well-known widely used diversity measures include; species richness, Shannon index, Simpson index, Pielou index, Margalef index, Brillouin index, Berger-Parker index and McIntosh U index. For the sake of this study, the species richness index, Shannon diversity index, Simpson diversity index and Pielou index will be considered (Fedor and Zvaríková 2019).

Species richness is a measure of the variability or variety of species based on a simple count of the unique number of species in a particular sample. From a more useful perspective, it can be expressed as species richness per unit area in a particular site or plot level referred to as alpha diversity or at the landscape scale species richness referred to as gamma diversity. The terms species diversity and richness are used interchangeably in some contexts to depict the different number of species in the area of interest, however, these might have different meanings and therefore should be used differently. The species diversity considers in addition to the number of species also the number of individuals and therefore it is prudent to distinguish

these terms. Despite the simplicity involved in computing the species richness, it is undisputedly influenced by sample size and plot size in the sampling study and therefore reports of these values should be complemented with the sampling effort. For example, it has been obvious that, the greater the sampling effort, the greater the possibility of a higher index for species richness explaining the tendency to explore new species with an increase in effort. The rarefaction curve of incidence or species accumulation curves are mostly used in studies to depict this and also extrapolate the required effort needed to achieve the required richness of the study area.

2.4.2 Shannon diversity

The index is a product of an expert mathematician and electrical engineer scientist Claude Shannon owing to his research which was geared towards measuring the amount of order or disorder within a certain system and further adopting it as part of links between information and diversity (Shannon & Weaver, 1949). The Shannon diversity index or Shannon – Wiener index, is commonly adopted in the field of ecology and ecological monitoring. The Shannon function is denoted as H' and is expressed as;

$$H' = -\sum_{i=1}^{S} pi\, log\ (pi)$$

where H' is the Shannon diversity index, S is the species richness, pi is the relative abundance of the *i*th species expressed as the abundance of the species to the total abundance and the natural logarithm is taken for the pi value (Spellerberg & Fedor, 2003). For ecological studies, the Shannon diversity is usually estimated at the plot level for all species in response to their relative abundance in the plot and further aggregation of plot values with their standard error reported at a certain confidence interval. The Shannon index results then in a single value, a measure of diversity in the direction of evenness or equitability. The index considers both the evenness and richness of the ecosystem. As much as this outcome is essential in determining the diversity status of an ecosystem, it is challenging to interpret especially for just one plot. Therefore, they are mostly recommended for comparative studies. The typical values of the Shannon index are generally in the range of 1.5 and 3.5 in most ecological research and rarely is the index greater than 4. Higher Shannon index values depict higher diversity in the study area. Therefore, they can be restructured that the Shannon index increases in the direction of increasing richness and evenness of species in the community. Shannon values of three and above can be achieved in tropically diverse ecosystems. An advantage of the Shannon index is

that it is not highly influenced by sample size and captures greater information in its expression (information on richness and evenness is represented). For example, for a community or ecosystem with knowledge of the species' equitability or evenness, the Shannon diversity is expected to express values more sensitive to species richness and vice versa (Fedor and Zvaríková 2019).

2.4.3 Simpson diversity

Simpson diversity index is attributed to the statistician Edward H. Simpson in his published work in 1949. Similar to the Shannon, the Simpson index has been adopted widely in the field of ecology to assess species diversity. The index is denoted by the symbol D and the expression is provided below;

$$D = \sum_{i=1}^{S} pi^2$$

The Simpson index specializes more in dominance due to its counts on common or dominating species. This then implies that a few rare species with corresponding few individuals representing them will not contribute remarkably or will have less effect on the overall value representing the index. The Simpson index can therefore be interpreted as the probability that any two individuals drawn from the sample at random belong to the same species. The index being a measure of diversity takes into account the number of species or richness and their respective abundance and is expected that, as species richness and evenness increase, so should the diversity index. The output value of the index ranges from 0 to 1, with 0 representing unlimited diversity and 1 indicating no diversity. If the possibility is that two randomly drawn species belong to the same species, then a higher index is expected. This implies that the concept of understanding values is directly opposite in the case of Shannon values. Lower values insinuate more diversity while higher values imply less diversity. Due to this interpretation and the ease of communication and presentation, the index is mostly expressed as a reverse in the case of equation I below or as a reciprocal in equation II below. In the case of reverse, the values are between 0 and 1 with higher values depicting higher diversity and lower values for less diversity. This then changes the narrative to the probability that two randomly selected individuals from the sample will belong to different species. The reciprocal produces values of D > 1 with higher values representing higher diversity and vice versa. There is no accepted way of presenting the index and it mostly depends on the researcher which is convenient to adopt in the study.

$D = 1 - \sum_{i=1}^{S} pi^2$ ……. Equation I

$D = \frac{1}{\sum_{i=1}^{S} pi^2}$ …….. Equation II

The Simpson index is highly influenced by the evenness of species distribution with species richness having comparatively weak influence.

The Shannon and Simpson indices are commonly used in parallel in ecological studies and this has been a general trend. The two indices are less affected by sample size and in most submissions regarded as generally negligible. The species richness and evenness concepts are captured fairly by the indices, notwithstanding, one concept is relatively more pronounced in dictating the output than the other. For example, the Shannon diversity index is strongly affected by rare and species richness and therefore is sensitive to small diversity variations which then is a better representation of the diversity status in the community, whereas the Simpson index favours more in the direction of common species and evenness and therefore a better depiction of species dominance in the community. From this perspective, it is commendable to adopt both indices in diversity studies, however, the choice strongly depends on the objectives of the study and what the emphasis is. Notwithstanding, a particular index cannot be overlooked in certain studies, for example, research with an interest in assessing recovery status after a disturbance will realize the importance of adopting the Shannon index due to its sensitivity or response to the return of initially rare species. For an ecosystem where the availability of certain species is of high relevance due to their commercial value or importance as a source of resource, their dominance is extremely preferable, and therefore the Simpson index cannot be overlooked in this context. For example, in the Budongo forest reserve where relevance is attached to mahogany species due to their incredible commercial value, studies in the pursuit of the economic status of the forest will be driven more in the direction of adopting the Simpson diversity index over the Shannon index. A contrast to this will be studies on biodiversity status after subjecting the forest to logging. A holistic approach of adopting both indices in a relevant perspective will be assessing the recovery status of the area after years of logging where there are multiple interests in the dominating species after these years and also the recovery capacity of all the species including the rare species (Fedor and Zvaríková 2019).

2.4.4 Pielou evenness index

The Pielou evenness index is based on the concept of equitability and as the name suggests, is a measure of species evenness in an ecological study. Evenness expresses the extent to which individual species are represented in the sample, expressing the distribution and relative abundances of species in the study. Two scenarios can be encountered when dealing with evenness, an extreme case where a single species dominates the ecosystem and the others present in meagre numbers (this will present the lowest equitability with index values close to 0) and a situation where all the available species are fairly or equally represented (expected highest equitability of nearly 1) (Fedor and Zvaríková 2019). To portray this in an example, for instance, in a sampling study of 5 species richness and total abundance of 50 for the individuals, the first case of lower equitability would be one species representing an abundance of 42 and the others of counts of 2 individuals each. The outcome may also be represented by 10 individuals each for the 5 species and this will then portray the second scenario of highest equitability of index of 1. This explains the range of Pielou evenness index values (from 0 to 1), with higher values indicating higher evenness or fair distribution in the abundance of the species present and lower values implying lower evenness with few species expected to be represented abundantly relative to the others. The Pielou evenness index is denoted by the letter J' and can be estimated from the mathematical algorithm below;

$$J' = \frac{H'}{ln(S)}$$

Where H' is the Shannon Wiener diversity index and S is the maximum level of species diversity possible within a forest community (Pielou, 1975). In ecological studies where Shannon and Simpson's diversities are already presented, the Pielou index is also recommended as a complement to these indices since it is focused specifically on species evenness providing a nuanced perspective of the community.

CHAPTER THREE

3.0 Methodology

3.1 Study area

3.1.1 Overview of Uganda's Forest Ecosystems

Uganda has a population of about 39 million with a total area of 241,559 km^2, out of which 24% representing approximately 5 million hectares is forested. Generally, Uganda has a tropical climate that has dictated and influenced the availability and growth of tropical rain forests and a vast portion representing the Eastern extension of the highly diverse Congo basin, home to a significant array of wildlife and tree species. Several primary forests have been gazetted as national parks and reserves harbouring extremely unique species. Among these are the; Bwindi Impenetrable National Park, Kibale Forest National Park, Budongo Forest reserve, Bugoma Forest Reserve, Mabira Forest, Mpanga Forest, Kalinzu -Maramagambo Forest, Mt Elgon Forest Reserve, Rwenzori Forest Reserve and Otzi Forest Reserve. Over the years, the country's increasing population has presented an obvious dilemma in resource management in the tropics; high-level pressure and overdependence on natural resources especially forestry to meet subsistence and commercial demand. Implemented policies and efforts from other environmental organisations have strived to manage forest resources to meet global contemporary challenges however sustainably, deforestation and forest degradation arising from illegal logging, agriculture, charcoal production and timber trade in the region have upsurge rendering all efforts futile. According to the reports from the WWF (World Wide Fund for Nature) in 2020 on forest resources in Uganda, it was estimated that the annual rate of forest loss in the country was approximately 122,000ha. This is of global concern considering the biodiversity-rich natural tropical ecosystems available in the country and its role in addressing global discourses. Dedicated effort in recent years has been directed towards this sector with potential and curbing it in the future. These include the nation's incorporation with the East African Community (EAC) Vision 2050 to be committed to realising the African Union Agenda 2063 and the Sustainable Development Goals (SDGs) with greater effort to ensure the reputation of their tropical forests. The signing of the Zanzibar declaration to address illegal logging and timber trade by protection agencies and further strengthen the policies on available protected forest areas is another commendable achievement (WWF, 2020).

3.1.2 The Budongo Forest Reserve

The study was conducted in Budongo Forest Reserve which forms the northernmost arc of the Albertine Rift forests in western Uganda, East Africa, with coordinates; 1°35´ to 1°55´N and 31°18´ to 31°42E´ at an altitude of 1,100m above mean sea level. The forest reserve is a tropical rainforest well known for its outstanding services; from a source of livelihood, climate regulation, biodiversity conservation and the largest composition of commercial species. The forest area has been subjected to various silvicultural activities until recent years. The initial activity was the selective removal of commercially valuable species after the chemical girdling or tree poisoning of undesirable species followed by the enrichment planting of mahogany species (Khaya anthotheca and Entandrophragma spp.). This is also termed as the arboricide treatment where tree species classified as 'weed species' are eliminated to open up the canopy. Though this activity was later abandoned to encourage natural regeneration, a continued activity would have resulted in a transformation of the species composition of the forest in the future where only a handful of commercially valuable timber dominated. The management of the reserve resulted in the division into eight forest blocks which are also subdivided into compartments to address the production, protection recreating and research management regimes. The National Forest Authority (NFA) is the government agency responsible for the management of the forest and issues licenses accompanied by management plans for operations in the production areas. Harvesting is prohibited in the strict nature reserve according to the NFA, however, illegal timber harvesting from pitsawyers persists in the reserve. According to (Babweteera et al., 2018), meeting the timber demands of the boat-making industry is another source of unsustainable harvesting in the reserve owing to the extinction of species like Cordia millenii coupled with its inability to establish under closed canopy. Conservation and development interventions have not been the priority of management in the past, however, a recent endeavour in the ecotourism and conservation project, the Budongo Conservation Field Station (BCFS) from the Budongo Forest project has contributed significantly to the sustainable management of the reserve. The conservation station is therefore actively promoting research to enhance knowledge about the status of the reserve especially after its past disturbance. Prime conservation areas in Uganda therefore cannot exclude the Budongo forest reserve.

The reserve has a total area of over 793 km^2 and is Uganda's largest forest reserve (Okimat et al., 2023). The study site is classified as a tropical moist deciduous forest with a highly diverse species composition located on severely weathered soils. It is estimated that approximately 260 species are located in this forest with Khaya anthotheca and cynometra alexandrii being the

most dominating species. The soils in the study area are highly weathered Lixisols characterized by high soil pH and base saturation. According to (Manu et al., 2022), soils here are higher in pH with potassium identified as a limiting nutrient for forest productivity. The area experiences a bimodal pattern of rainfall of an annual average of 1670mm with an average air temperature of 22.8 degrees Celsius (Manu, 2022). Tree species of the family Meliaceae which are commercially valuable were mainly considered during the selective logging in the study site with recorded logging accompanied by set management plans commencing way back in 1945. A portion of the area was designated from the onset of logging as a nature reserve which has received strict protection regulations. The area after selective logging has been designated as a research area and therefore activities have been prohibited for over 70 years. The Budongo forest consists therefore of an unlogged area and a selectively logged area. This study was carried out in the unlogged forest that occurs in compartment N15 and a selectively logged forest that occurs in compartment N3. Compartment and forest types will be used interchangeably for logged and unlogged forests.

3.2 Sampling and Data Collection Methods

Data was collected from compartments N3 (70 years) (logged) after selective logging and a primary forest (compartment N15)(unlogged) which has not been logged and without information about the exact age. According to Plumptre, 1996, compartment N3 was logged from 1947 to 1952 with approximately 80 m^3/ha of total timber volume removed with Mahogany accounting for 39.1 m^3/ha. Both compartments are located in the Nyakafunjo block with the total area of N3 being 620 ha and N15 as 777 ha. Inventory was carried out in systematically established permanent plots of 204 circular plots of 1,000 m^2 at 100m intervals along a 2.5 km transect, with each nested with four 4 m^2 subplots in selectively logged and unlogged forests in the study area for phenology monitoring. In total, 204 main plots (105 in logged and 99 in unlogged) and 816 subplots (420 in logged and 396 in unlogged forest) (Okimat et al., 2023). Tree species with DBH greater or equal to 10cm were identified and measured in the 1,000 m^2 plots and this accounted for the canopy layer of the study. Enumeration of natural regeneration in the subplots (4×4 m^2) was done for species with DBH less than 10cm (seedlings and saplings) and this was classified as the regeneration layer for this study. Diameter at breast height (DBH), Height (H), canopy closure, tree coordinates and names of species were recorded. Trees were identified to the species levels on the field and also with the use of the institutional herbarium database. Soil samples were collected from each

of the subplots and composited and analysis was carried out, with the results of each composited soil representing the entire plot. Samples were collected from the upper 20 cm of the mineral soil using a 7 cm diameter extracting auger. For the sake of this study, interest was in the results for soil pH, Sodium (Na), Calcium (Ca), Iron (Fe), Cation exchange capacity (CEC), Aluminium (Al), Magnesium (Mg) and Potassium (K). For a complete explanation of the sampling and design, refer to (Okimat et al., 2023).

3.3 Analysis

This study assumes that the compartments (selectively logged forest) had similar structural and species composition before it was subjected to logging and therefore any discrepancy or variance is directly attributed to the impact of logging. To assess tree species composition and diversity between the logged and unlogged forest, the analysis was conducted independently at the sampled plots (n = 204) level each for the logged and unlogged forest and between the two forest types. Local species diversity was determined by comparing alpha (α) beta (β) and gamma diversity ($\gamma = \alpha \times \beta$) between the logged and unlogged forests. Alpha diversity is the measure of total number of species at the plot level. The diversity between plots was determined by the beta diversity and the gamma diversity for the landscape scale. The interpolation and extrapolation rarefaction curves were developed as a measure of diversity from the sampling effort between logged and unlogged forests for the canopy and regeneration layer from species richness and their corresponding abundances. The rarefaction determines the number of expected species in a randomly selected and indicates whether an increase or discovery of new species can be expected with additional sampling effort. Rarefaction also presents to some extent the sampling study's performance and meticulousness in assessing richness in the study area. The rarefaction curve is expected to culminate as an indication of asymptote and a level of confidence in the available species sampled for the study and a fair indication of what to expect when more effort is invested beyond the current outcome. This was achieved by using the iNEXT package in R (Hsieh et al., 2016). This was followed by comparing the alpha diversity between the forest types at the canopy and regeneration layer. Differences in plot sizes were accounted for by adopting the rarefaction approach of calculating standardized values from the estimated alpha diversity for the canopy (1000m^2) and regeneration (16m^2) layers. This was used to generate a 1:1 graph between the canopy and regeneration layer to compare the differences in alpha diversity between the forest and also the forest types. Therefore, the scatterplot presented is meant to portray the contrast in alpha diversity between

the canopy and regeneration layer for all the plots. The diagonal 1:1 line presents a slope of 1 indicating equal unit increase for both the X (Canopy layer) and Y (Regeneration layer) axis. With this established, any point that falls below the line implies that the canopy layer in that plot is richer or more diverse than the regeneration layer and vice versa, with points on the diagonal line signifying equal diversity for both layers.

The following diversity indices were estimated as well using the Vegan package in R (Oksanen Jari et al., 2024). Species richness(**S**), Simpson's index (**D**) - ($D = \sum_{i=1}^{S} pi^2$), Shannon index(H') – ($H' = -\sum_{i=1}^{S} pi\ log\ (pi)$) and Pielou's evenness index (J) – ($J = \frac{H'}{H'_{max}}$). The mean, standard error (SE) and analysis of variance (ANOVA) were also estimated for the forest types at the regeneration and canopy layers and p-values reported at a 95% confidence interval. To examine the recovery of the selectively logged forest relative to the unlogged forest, the basal area, above-ground biomass (AGB), canopy cover (percentage closure) and density of tree species were determined in addition to their statistics. Taxonomic classification to the species level was done for each identified tree. Species were then categorized into diameter size classes; seedlings and saplings of DBH < 10cm representing the regeneration layer and trees of DBH ≥ 10cm representing the canopy layer for both logged and unlogged forest (Hall et al., 2003). The differences in the aboveground biomass, stem density and basal area between the selectively logged forest and unlogged forest were tested with analysis of variance (ANOVA) to determine if the change is statistically significant. Family of species, species ecological guild (shade bearers, Non-pioneer light demanders (NPLD) and pioneers), deciduousness and wood density were presented in a tabular form for both logged and unlogged forests. To assess the relative recovery, the importance value index (IVI) was estimated to present the most important species and ascertain the extent of dominance. This was estimated from the equation:

$IVI = \frac{RD+RDo+RF}{3} \times 100\%$: where RD is the relative density = (Number of individuals of a species) / (Total number of individuals in the community), RDo is relative dominance = (Basal area of a species) / (Total basal area in the community) and RF is relative frequency = (Number of plots where a species is observed) / (Total number of survey plots). Due to the absence of incomplete measurements for the regeneration layer, the IVI was determined for tree species at the canopy layer (DBH ≥ 10cm). Pantropical allometric models from (Chave et al., 2005) were used for the estimation of AGB as illustrated below:

$AGB = 0.0673 \times (\rho D^2 H)^{0.976}$ ……1

$$AGB = \rho \times e^{(-1.499+2.148\, lnln\,(D\,)+0.207\,(lnln\,(D\,))^2-0.0281\,(lnln\,(D\,))^3)} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots 2$$

Where, ρ is the wood density, e is exponential function, D is DBH and H is height.

For the comparison of the similarity of species composition, differences in the species composition in the canopy and regeneration layer for the forest types were determined using the Non-metric multidimensional scaling (NMDS) ordinations for all the plots. The metaMDS under the Vegan package was used and the ellipses at 95% confidence level for multivariate t-distribution were fitted to clearly distinguish the various groups. The choice of NMDS over other ordinations is based on its performance in similarity analysis in the ordination space and its unique property of not assuming a linear relationship among variables like in the case of principal component analysis and principal coordinate analysis. The permutational multivariate analysis of variance (PERMANOVA) analysis was followed to statistically determine whether the differences between the groups were significant. Just as the ellipse portrays a visual representation of the similarity between and within the data points, the PERMANOVA estimate the significance between the groups as a result of whether the centroids (multivariate means) of the groups are different and also the variance or variability among the data points. The PERMANOVA is developed based on a null hypothesis that there are no differences in the species composition between the logged and unlogged forest. The Adonis and Pairwise Adonis under the *Vegan* and *devtools* packages were adopted for the PERMANOVA and pairwise PERMANOVA respectively (for forest type and forest layer).

To determine the influence of soil properties on the spatial distribution of tree species, the soil nutrients considered were fitted to the NMDS ordination for the logged and unlogged forests. Before this was done, a multicollinearity analysis was performed using the *car* package in R to estimate the Variance Inflation Factor (VIF) as a basis to retain or exclude soil variables for the ordination analysis. Based on this, CEC was excluded as it was highly collinear to Ca. Fe was excluded from the ordination due to inadequate variability explained in the data. The ordination was achieved by first performing the NMDS based on the species composition(plots) and also the species. The *envfit* function under the *Vegan* package was adopted to generate the soil nutrient vector which was then fitted to the NMDS plot. The target is to explore the magnitude to which these soil variables influence the distribution relative to K and pH by plotting them in one ordination space. This was achieved by fitting soil nutrient vectors from computing the mean values of the composited subplots to the NMDS ordination of the species and plots. In this context, the distance matrix of the plots was estimated based

on the species composition from the abundance of the individual species. The orientation and length of the arrows represented in the soil variables portray the relationship between the soil nutrients and the species and plots. For all NMDS ordinations, the default Bray Curtis dissimilarity and scaling were used. The magnitude and direction of influence of the soil properties within the logged and unlogged forests were then illustrated through the length and orientation of the fitted arrows and the means and SE were also determined and presented in a table. For all statistical tests, significant levels of $p < 0.05$ were used. Data organization and analysis were done using R software version 4.3.3.

CHAPTER FOUR

4.0 RESULTS

4.1 Tree Species Composition, Diversity and Recovery

4.1.1 Rarefaction curves of Canopy and Regeneration layers for logged and unlogged forests

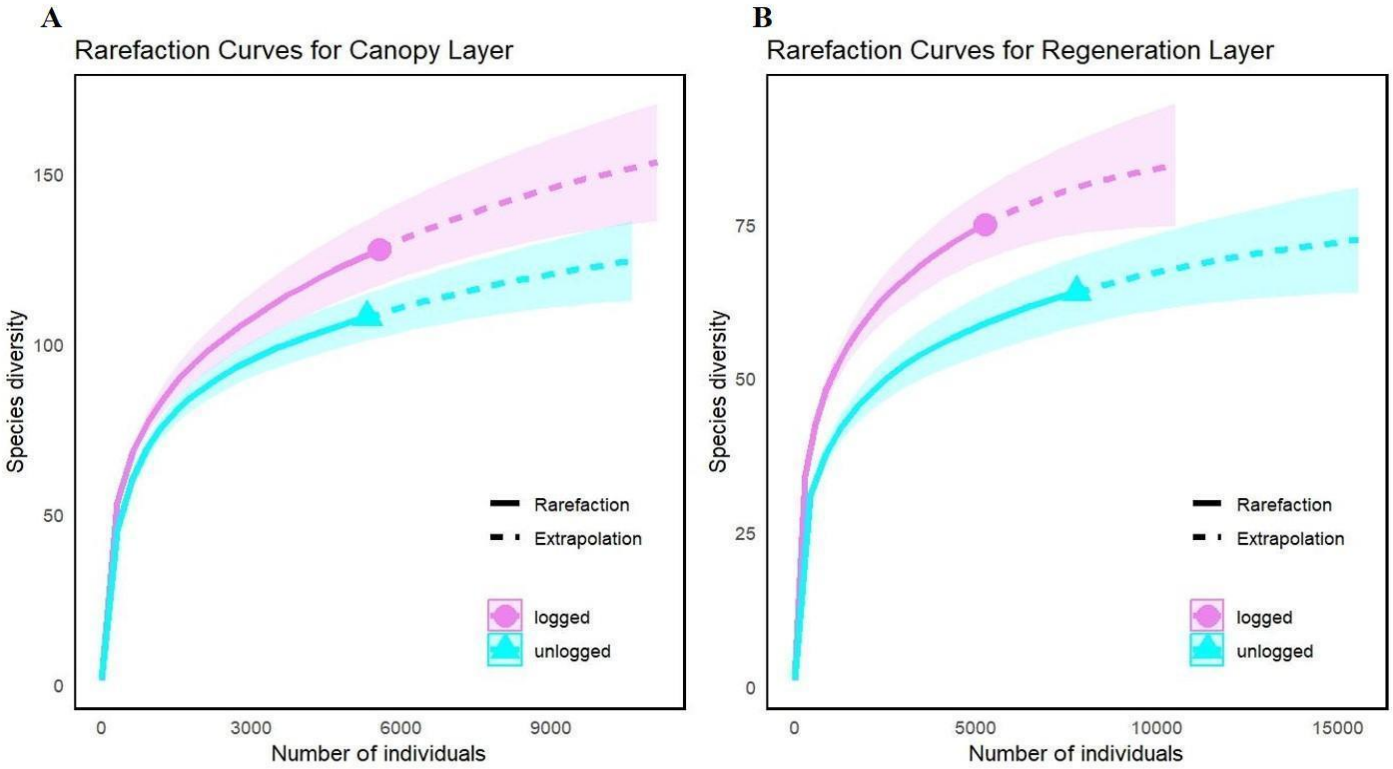

Figure 1: Rarefaction curves of species richness and abundance with 95% confidence intervals. Solid lines represent observed and interpolated species diversity, while dotted lines represent predicted species diversity based on extrapolation. Canopy and Regeneration curves are based on sample plots of 1000m^2 and 16m^2 respectively.

Rarefaction curves are developed based on rarefaction analysis of species richness and abundance which endeavour to portray the completeness of a sampling study and sampling effort. The rarefaction curves for the canopy layer (A) and regeneration layer (B) exhibit a common pattern. At the commencement of the sampling study, since no species have been recorded, an initial establishment of the plot yielded a drastic increase in the species diversity since almost every observed species is likely to be recorded as a new species. For the rarefaction curves for the canopy layer (A), an initial blistering increase in species diversity is observed and between 100 and 150 species, the curves are rather steady however they are not at the point of culmination. Depending on the trend, extrapolations are developed to portray the expected required increase in sample size or effort to appreciatively represent the study area. In both the canopy and regeneration layer, it is portrayed both in the observed or interpolation and extrapolated that, the species diversity and abundance in the logged are higher

and expected to comparatively increase. In both situations, the logged forest is richer in species composition than the unlogged forest with the extrapolation predicting even higher species to be discovered. A total of 128 species representing an abundance of 5560 were recorded for the logged forest canopy layer (A) with the unlogged recording 108 species and an abundance of 5312. A total of 154 unique species were identified at the canopy layer representing a total abundance of 10872, with 46 species exclusive to logged and 26 for unlogged forest. The extrapolation curves for the unlogged forests are steadier than the logged illustrating the possibility of exploring or identifying more species with increasing sampling effort for the logged and the near climax of identifying new species for the unlogged forest. At the regeneration layer (B), a total of 87 unique species representing 13047 individuals were recorded. A total of 72 and 65 species were observed in the logged and unlogged forest representing 5257 and 7790 respectively. The logged regeneration layer had exclusively 22 species that were absent in the unlogged regeneration layer and 15 species were exclusive to the unlogged regeneration layer. The trend portrayed is expected because subjecting the forest to disturbance through logging will enable conditions likely to elevate the species diversity. Though it can be visualized that Figure 1A is richer in species diversity than Figure 1B, this conclusion cannot be justified and is not advisable since the plot sizes were different for the canopy and regeneration layers (1000m^2 and 16m^2 for A and B respectively).

4.1.2 Comparing Alpha diversity (plot species richness) between the Canopy and Regeneration layer at the forest level and type (Standardized Alpha diversity)

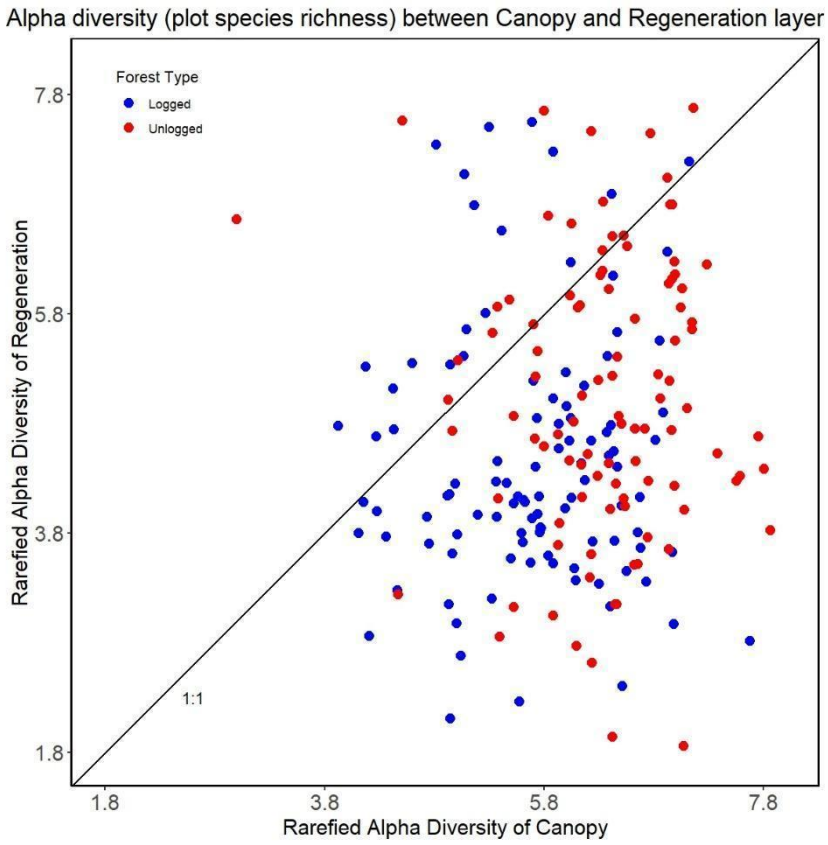

Figure 2: Rarefied alpha diversity between Canopy and Regeneration layer. The rarefied alpha diversity values are a measure of standardized species richness values for each plot (alpha diversity) and the diagonal 1:1 line depicts a slope of 1. A total of 204 sample plots for logged and unlogged forests were adopted.

The figure above directly compares the species richness at the plot level (alpha diversity) between the canopy (overstory) layer and the regeneration (understory) layer after accounting for differences in plot sizes. The first comparison is at the forest level; canopy and regeneration layer (without any colour segregation), and the second is a comparison of the canopy and regeneration layer at the forest type level; logged (blue coloured points) and unlogged (red coloured points). At each comparison, there are more points with diversity values in favour of the canopy layer depicting that most of the sample plots had species diversity higher in the canopy layer than the regeneration layer. Aside from this trend, the unlogged forest relatively had more points closer to or lying on the 1:1 line at the forest-type level, indicating that sample plots that had species richness nearly or equal in value at the different layers are expected to be found in the unlogged forest than the logged forest. Evaluating this with the values estimated for the sample plots for the alpha diversity, the mean values were higher in the canopy layer

than the regeneration layer as already illustrated and the p-values all returned these differences as being statistically significant (Table 1). Based on Figure 2 and Table 1, it can be inferred that, after accounting for differences in plot size (area), a diversity difference is still observed with the canopy layer demonstrating higher diversity than the regeneration layer.

Table 1: Comparing the differences in species richness between canopy and regeneration layer for the forest types (logged and unlogged) and entire forest (combined forest) from standardized alpha diversity values. The means and standard errors (SE) are presented with F and p-values generated from Analysis of Variance (ANOVA) results at a confidence interval of 95% (0.05).

Variable	*Category*	*Canopy layer mean (SE)*	*Regeneration layer mean (SE)*	*Sig.*
Rarefied Alpha Diversity	Logged forest	5.70(0.08)	4.42(0.12)	***
	Unlogged forest	6.36(0.08)	5.02(0.13)	***
	Combined Forest	6.023(0.06)	4.71(0.09)	***

*** indicate significant level at $p<0.001$.

4.1.3 Determining Tree species diversity and indices between the forest types or compartments (logged and unlogged)

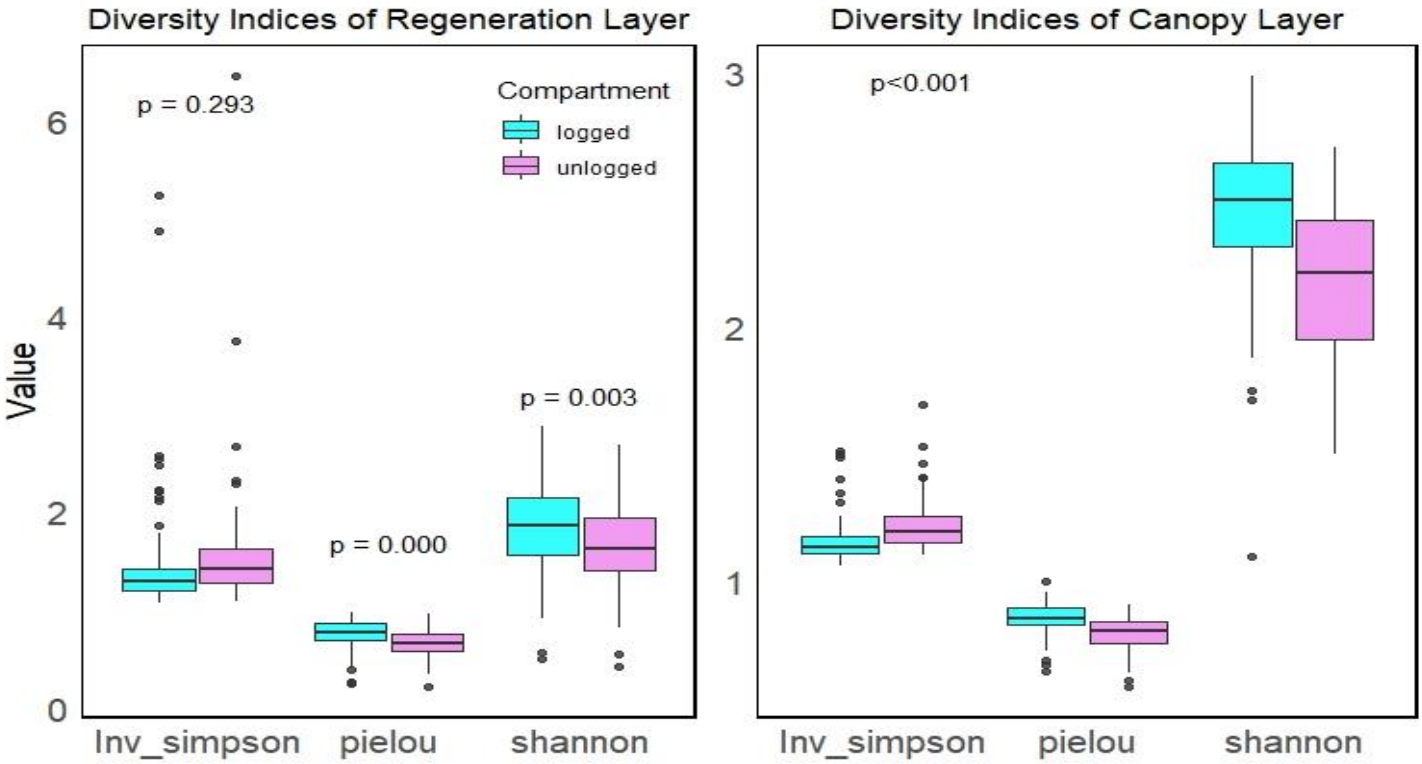

Figure 3: Diversity indices for regeneration and canopy layers between logged (cyan colour) and unlogged (purple colour) forest. Comparison of Pielou evenness, Shannon and Inverse Simpson diversity indices. P-value generated from ANOVA for plot means at a confidence interval of 95%. The plot presents the variability in the dataset for regeneration and Canopy layers based on 205 and 204 sample plots of $16m^2$ and $1000m^2$ respectively.

The species diversity(richness) and abundance (evenness) are the main determinants of the diversity index output. The total species richness (gamma diversity) for the regeneration and canopy layer in the study area were 87 and 154 respectively. An assessment of the species richness between these two forest types indicates a higher diversity in the logged forest than in the unlogged forest for both regeneration and canopy layers. For the species observed, a total species abundance of 13047 was recorded for the regeneration layer with the logged and unlogged representing 40% and 60% respectively. A total abundance of 10872 was recorded for the canopy layer with the logged and unlogged representing 51% and 49% respectively. The total abundance is relatively higher for lower species richness in the unlogged forest. The Pielou evenness index is therefore essential to ascertain the equitability or dominance in the study area. For the Pielou index, at the regeneration layer, the median estimated score was 0.771 (with an interquartile range (IQR) of 0.186 extending from 0.682 to 0.868) and 0.666 (IQR of 0.169 extending from 0.584 and 0.753) for logged and unlogged forests respectively.

A mean value of 0.75 ± 0.015 and 0.67 ± 0.014 was recorded for the logged and unlogged forest (Table 2). At the canopy layer, the median estimated score was 0.854 (with an interquartile range (IQR) of 0.07 extending from 0.825 to 0.893) and 0.805 (IQR of 0.08 extending from 0.756 to 0.840) for logged and unlogged forests respectively. A mean value of 0.85 ± 0.006 and 0.79 ± 0.007 was recorded for the logged and unlogged forest (Table 2). In each situation, the Pielou evenness index was higher in the logged compared to the unlogged forest with the differences in means being statistically significant (Table 2).

The Shannon diversity index is a measure of the diversity status of the study area. At the regeneration layer, the estimated median was 1.859 (IQR of 0.597 extending from 1.559 to 2.156) and 1.626 (IQR of 0.54 extending from 1.408 to 1.948) for logged and unlogged forests respectively. A mean value of 1.84 ± 0.045 and 1.66 ± 0.043 was recorded for the logged and unlogged forest (Table 2). At the canopy layer, the median estimated score was 2.50 (IQR of 0.33 extending from 2.316 to 2.645) and 2.212 (IQR of 0.47 extending from 1.949 to 2.423) for logged and unlogged forests respectively. A mean value of 2.47 ± 0.028 and 2.18 ± 0.03 was recorded for the logged and unlogged forest (Table 2). Relatively, the variability was higher in the logged forest than in the unlogged with the regeneration layer exhibiting a similar trend explained by the outliers. For both forest layers, the Shannon diversity index was higher in the logged forest than in the unlogged denoting higher species diversity in that regard, with the differences in means for the regeneration layer and canopy layer determined to be statistically significant (Table 2). The Simpson and Inverse Simpson diversity indices also measure species diversity but they are specialised more towards measuring dominance. For enhanced presentation and interpretation of results, the Inverse Simpson is adopted which presents higher values for higher diversity and vice versa. The median inverse Simpson for the regeneration layer was 1.288 and 1.417 with means of 1.45 ± 0.06 and 1.54 ± 0.06 for the logged and unlogged forest respectively. In the canopy layer, it was 1.135 and 1.196 with means of 1.15 ± 0.008 and 1.22 ± 0.01 for the logged and unlogged forest respectively. The Inverse Simpson is therefore higher in the unlogged forest than the logged forest. High variability in the plot values were observed, portrayed by the outliers. The driving factor here is how the forest diversity indices have been influenced owing to the introduction of selective logging to the forest at both the regeneration and canopy layers. Therefore, based on Figure 3 and Table 2, the following inferences are highlighted; With regards to gamma diversity, the logged forest is more diverse than the unlogged, Pielou and Shannon diversity indices were also higher in the logged than the unlogged forest with this further portrayed in the higher

abundance of the unlogged forest. Inverse Simpson was higher in the unlogged forest but more pronounced in the canopy than in the regeneration layer.

Table 2: Statistics of the diversity indices of regeneration and canopy layers for logged and unlogged compartments within the Budongo Forest Reserve, Uganda. The mean and standard error (SE) are presented with F and p-values generated from Analysis of Variance (ANOVA) results at a confidence interval of 95% (0.05).

Category	*Variable*	*Logged forest mean (SE)*	*Unlogged forest mean (SE)*	*F value*	*P value*
Regeneration layer (DBH < 10cm)	Shannon index	1.84 ± 0.045	1.66 ± 0.043	8.931	0.0032
	Exponential Shannon	6.95 ± 0.29	5.74 ± 0.257	9.627	0.0022
	Simpson index	0.74 ± 0.014	0.69 ± 0.013	6.281	0.013
	Inverse Simpson	1.45 ± 0.058	1.54 ± 0.061	1.111	0.293
	Pielou index	0.75 ± 0.015	0.67 ± 0.014	15.07	0.00014
	Alpha diversity(α)	11.71 ± 0.268	12.14 ± 0.311	1.129	0.289
	Trees (N/m^2)	3.13 ± 0.17	4.869 ± 0.285	28.14	<0.000
	Gamma diversity(γ)	72	65	-	-
Canopy Layer (DBH ≥ 10cm)	Shannon index	2.47 ± 0.028	2.18 ± 0.03	49.35	<0.000
	Exponential Shannon	12.23 ± 0.302	9.24 ± 0.262	55.53	<0.000
	Simpson index	0.87 ± 0.005	0.83 ± 0.006	33.67	<0.000
	Inverse Simpson	1.15 ± 0.008	1.22 ± 0.01	26.97	<0.000
	Pielou index	0.85 ± 0.006	0.79 ± 0.007	44.21	<0.000
	Alpha diversity(α)	18.51 ± 0.359	15.85 ± 0.343	28.7	<0.000
	Trees (N/ha)	529.5 ± 13.60	536.6 ± 9.44	0.177	0.674
	Basal area(m^2/ha)	37.27 ± 1.319	39.87 ± 1.846	1.335	0.249
	AGB(kg/ha)	455.63 ± 23.48	576.95 ± 36.83	7.899	0.005
	Gamma diversity(γ)	128	108	-	-

4.1.4 Important Value Index (IVI)

The Important Value Index (IVI) measures the importance of individual species in the community, estimated from the sum of percentages of relative density, frequency and dominance. In this regard, species with high dominance and importance in the community are expected to portray higher IVI values. Eight of the ten most important species presented were shared between the logged and unlogged forest with *Celtis durandii* and *Maesopsis eminii* exclusive to the logged and *Rinorea ardisiaeflora* and *Bosqueia phoberos* to the unlogged (Table 3). In the unlogged forest, *Cynometra alexandri* (36.3), *Celtis mildbraedii* (35.4), and *Lasiodiscus mildbreadii* (32.5) were superior in their importance and dominance with the next species after them (*Rinorea ardisiaeflora*) recording half the value of the third IVI (17.5). In

contrast, the logged forest also recorded three species (*Funtumia elastica* (27.8), *Celtis mildbraedii* (26.8) and *Cynometra alexandri* (22.8)) with distinguishably the highest importance among the ten, however, unlike the unlogged forest, the difference after the third species is comparatively marginal (*Lasiodiscus mildbreadii* (14.6)). This illustrates a certain level of dominance in the unlogged forest relative to the logged forest. Previously highlighted in the Pielou evenness index that the logged forest is comparatively even than the unlogged forest (Table 2). Interestingly, the differences in the basal area which was a determining factor for the relative dominance were not statistically significant between the logged (37.27 ± 1.319) and unlogged (39.87 ± 1.846)) forests ($F = 1.335$, $p = 0.249$).

Table 3: IVI for the first Ten (10) species of the Logged and Unlogged forest arranged in the order from higher to lower. The percentages are estimated as relative values from the compartment and correspond to relative frequency, density and dominance. The IVI is a sum of the three percentages.

Species	Frequency (%)	Density (%)	Dominance (%)	IVI
		Logged Forest		
*Funtumia elastica (*FUE*)*	2.38	17.1	8.29	27.8
*Celtis mildbraedii (*CMI*)*	2.28	15.9	8.60	26.8
*Cynometra alexandri (*CYA*)*	2.48	5.40	14.9	22.8
*Lasiodiscus mildbreadii (*LM*)*	2.38	8.85	3.41	14.6
*Celtis zenkeri (*CZE*)*	2.48	5.56	4.67	12.7
*Khaya anthotheca (*KA*)*	2.38	2.21	6.70	11.3
*Celtis durandii (*CDU*)*	2.18	2.46	4.10	8.7
*Trichilia rubescens (*TRR*)*	2.28	4.41	1.85	8.5
*Maesopsis eminii (*ME*)*	1.58	0.88	5.39	7.86
*Alstonia boonei (*AB*)*	2.08	0.94	3.66	6.68
		Unlogged Forest		
*Cynometra alexandri (*CYA*)*	2.61	4.73	28.9	36.3
*Celtis mildbraedii (*CMI*)*	2.61	20.9	11.8	35.4
*Lasiodiscus mildbreadii (*LM)	2.61	21.8	8.03	32.5
*Rinorea ardisiaeflora (*RI*)*	2.61	11.9	2.99	17.5
*Trichilia rubescens (*TRR*)*	2.61	5.61	4.25	12.5
*Funtumia elastica (*FUE*)*	2.61	5.37	3.54	11.5
*Alstonia boonei (*AB)	2.48	1.21	4.88	8.57
*Celtis zenkeri (*CZE*)*	2.61	2.45	2.33	7.39
*Khaya anthotheca (*KA*)*	2.61	1.41	2.85	6.87
*Bosqueia phoberos (*BP*)*	2.61	1.45	1.69	5.75

4.1.5 Ecological guild of tree species

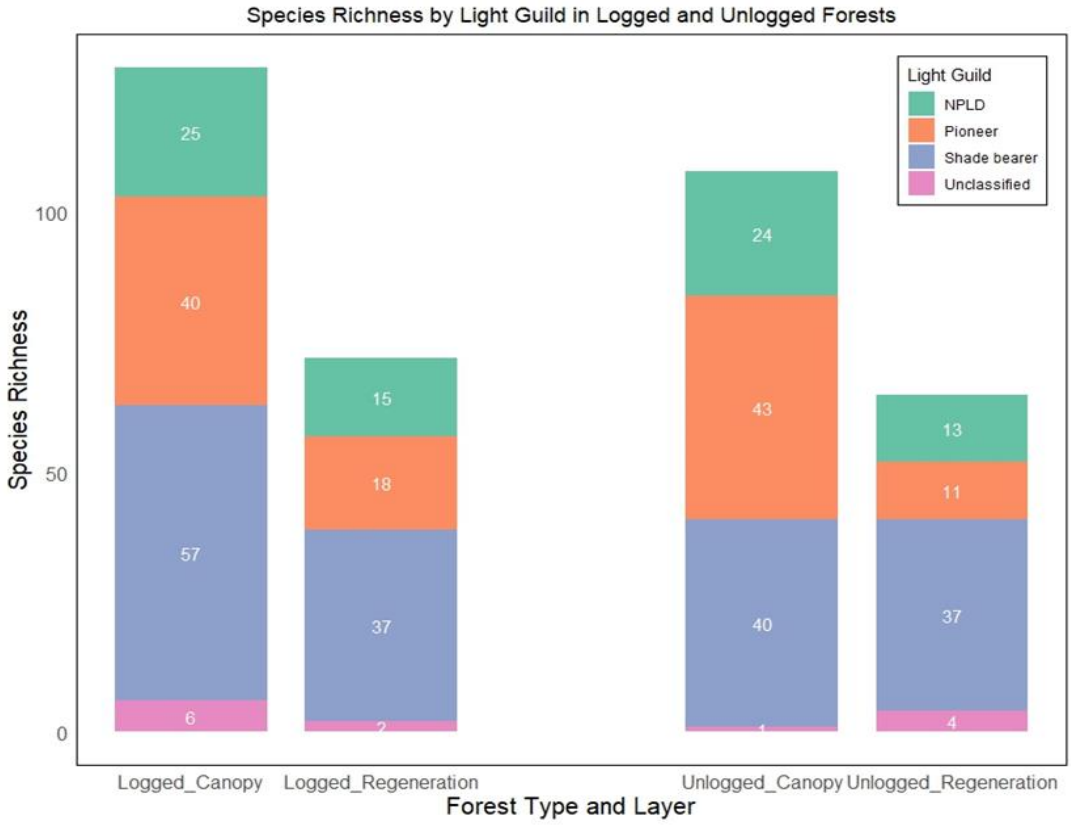

Figure 4: Differences in species richness(S) between the logged and unlogged forest for the canopy and regeneration layers and ecological guilds of the recorded species in the various categories. The number of unique species of each light guild is shown in the colour category. NPLD: Non-pioneer Light demanders. The unclassified are the group of species that were not able to be identified and were classified as unknown in the species category.

It is portrayed in the figure that shade bearers dominate the regeneration (57%) of the unlogged forest compared to the canopy layer (37%) where there is a fair representation of the various categories of species. A greater decrease of 50% is observed (from 40% in the canopy to 20% in the regeneration) in the pioneer species in the unlogged forest. This is attributed to the limited availability of light due to canopy establishment, promoting species tolerant to shade. This is depicted in the disparity and possible autocorrelation in species composition of the plots and therefore the expected high similarity of species composition due to the dominance of one class of species. A similar trend is portrayed at the regeneration layer of the logged forest as well, illustrating to some extent similar conditions established in the unlogged forest which then indicates that, the canopy layer is also well established in this forest type. Therefore, the trend expected is that the openness from logging induced the coexistence and growth of this remnant and the established cohort before the disturbance to grow into the new canopy and influenced the restructuring of the regeneration layer. This is evidenced that after 70years of logging, the forest has assumed a greater proportion of the recovery.

4.2 Spatial distribution of tree species and assessment of similarity

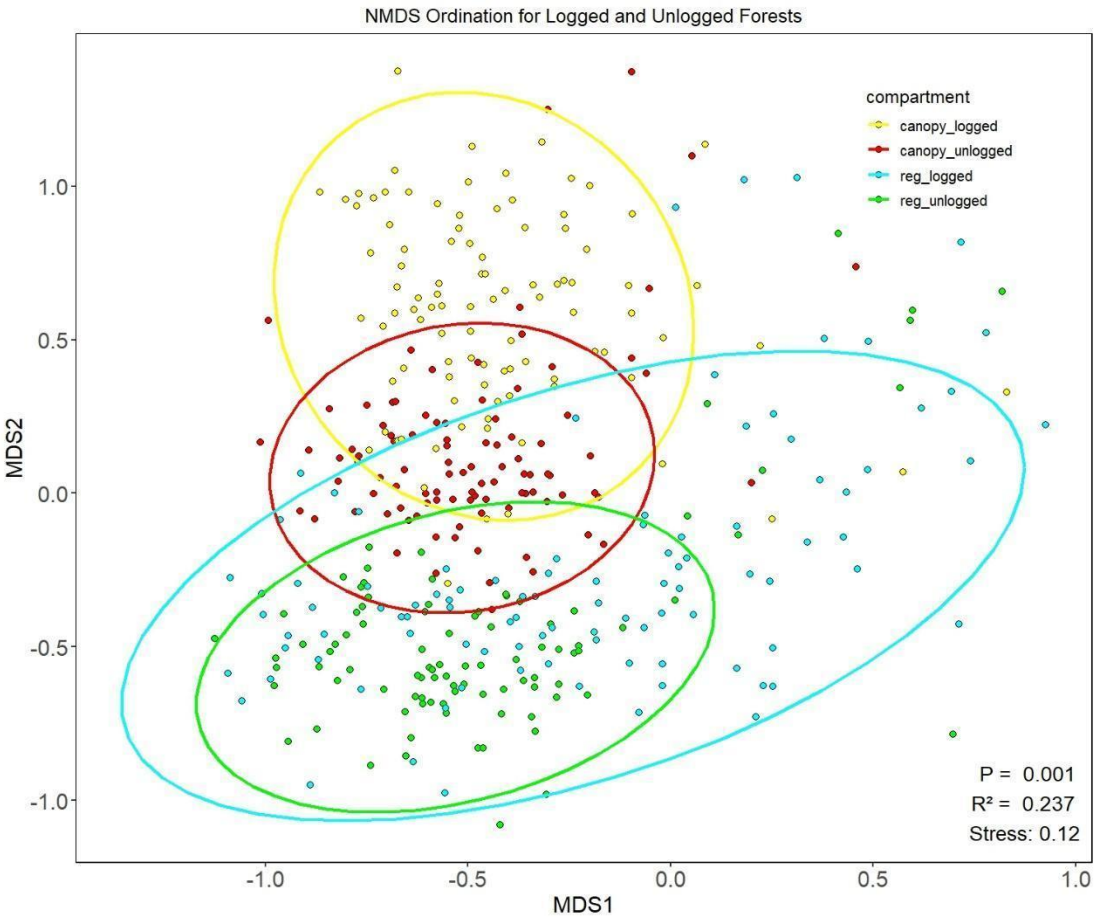

Figure 5: Non-metric multidimensional Scaling (NMDS) ordination for the forest types and layers based on the sample plots. The ellipses are developed at a 95% confidence level for a multivariate t-distribution with overlapping areas illustrating similarity. Regeneration is represented as reg in the legend.

The relative dissimilarity based on species composition was evaluated between the forest types (logged and unlogged) for each forest layer (canopy layer and regeneration layer) by developing a multidimensional ordination where each point represents the plots scaled based on the distance matrix and standardized species composition. The driving factor in the ordination is the discrepancy in species composition driven by logging, explaining the distinctiveness and therefore referred to as the beta diversity.

The NMDS figure illustrates that the logged forest features different but similar species composition with the unlogged forest for both the canopy and regeneration layer. With all the forest types and layers combined in one ordination space, the trend is well illustrated with pronounced clustering of points in areas more similar in species composition. The ordination provided a good stress value of 0.12. The proportion of variance explained for the entire data was 0.237 indicating approximately 24% of the variance explicated by the discrepancies. The

distinctiveness and similarities in the data points are well illustrated for the compartment types, with the PERMANOVA results revealing a statistically significant difference in the multivariate composition among the compartments ($F = 41.8$, $p = 0.001$) (Table 4). From the pairwise comparison, though the values representing the proportion of variance explained (R^2) was small, and the ellipses were overlapping in some comparisons, there was a statistically significant difference between all the pairs, elucidating the contrast in species composition.

It can be deduced that, the logged forest (regeneration and canopy layer) illustrates the extremes of the species composition while the unlogged forest (canopy and regeneration layer) comprising plots with species composition similar or incorporated in portions of the logged forest composition. It is worth highlighting that marginalized points from the logged forest represent the similarity to the unlogged with numerous points from the logged forest exhibiting uniqueness. A unique spectrum of species composition is engulfed within the logged regeneration layer encompassing the entire unlogged regeneration and a substantial portion of the unlogged canopy layer indicating a higher similarity in plot species composition between these compartment types. Among the compartment types, the logged forest (canopy and regeneration layer) demonstrated a high level of uniqueness with the points of the unlogged forest either represented in the canopy or regeneration layer of the logged forest. An intersection of species composition was observed for the unlogged canopy layer where the points are represented or overlapping with other compartment types as well. At the regeneration level between the forest types, aside from the similarity of a fragment of the logged with the entire unlogged, a greater portion of the points are unique. Just as all the points from unlogged are represented by the logged, not all points in the logged are similar to the unlogged. Furthermore, the unlogged regeneration layer has a species composition completely dissimilar from the logged canopy layer. It is interesting to note that the trend illustrated by the regeneration layers for the forest types was not emulated by the canopy layers. Therefore, it can be inferred from the figure that, after these years of subjecting the forest to disturbance, the initial species composition at the regeneration layer is maintained or still available but rather elevated with more species composition, the canopy layer has experienced drastic dissimilarity in species composition though a portion still represents a greater magnitude of the species composition in the absence of the disturbance. Few outliers were observed and it was fairly represented for all the levels considered.

In each situation, the logged forest is more diversified with a portion of its plots representing to a large extent a similar composition to the unlogged. Based on this, the following are highlighted, At the forest type (compartment) level, the logged forest encompasses the species

composition of the unlogged with numerous unique dissimilar plots. At the forest layer level, the logged canopy layer exhibited a pronounced distinction with all other layers similar in species composition to the regeneration layer of the logged forest. Similar to the logged forest, the unlogged forest also possesses species composition at the regeneration layer highly dissimilar from the canopy layer with few plots overlapping.

Table 4: Pairwise result from PERMANOVA for the different forest types and forest layers, F and p-values generated from Analysis of Variance (ANOVA) at a confidence interval of 95% (0.05). The variation explained in represented by the R^2.

Compartment Type	*Degree of freedom*	*Sum of Squares*	*F - value*	R^2	*P – value*
Regeneration unlogged vs Canopy unlogged	1	7.98	49.5	0.20	0.001
Regeneration unlogged vs Regeneration logged	1	2.91	14.3	0.07	0.001
Regeneration unlogged vs Canopy logged	1	15.08	82.4	0.29	0.001
Canopy unlogged vs Regeneration logged	1	5.76	30.2	0.13	0.001
Canopy unlogged vs Canopy logged	1	4.72	27.7	0.12	0.001
Regeneration logged vs Canopy logged	1	10.32	48.9	0.19	0.001
All compartments	3	23.41	41.8	0.24	0.001

4.3 Influence of soil properties on the spatial distribution and composition of tree species

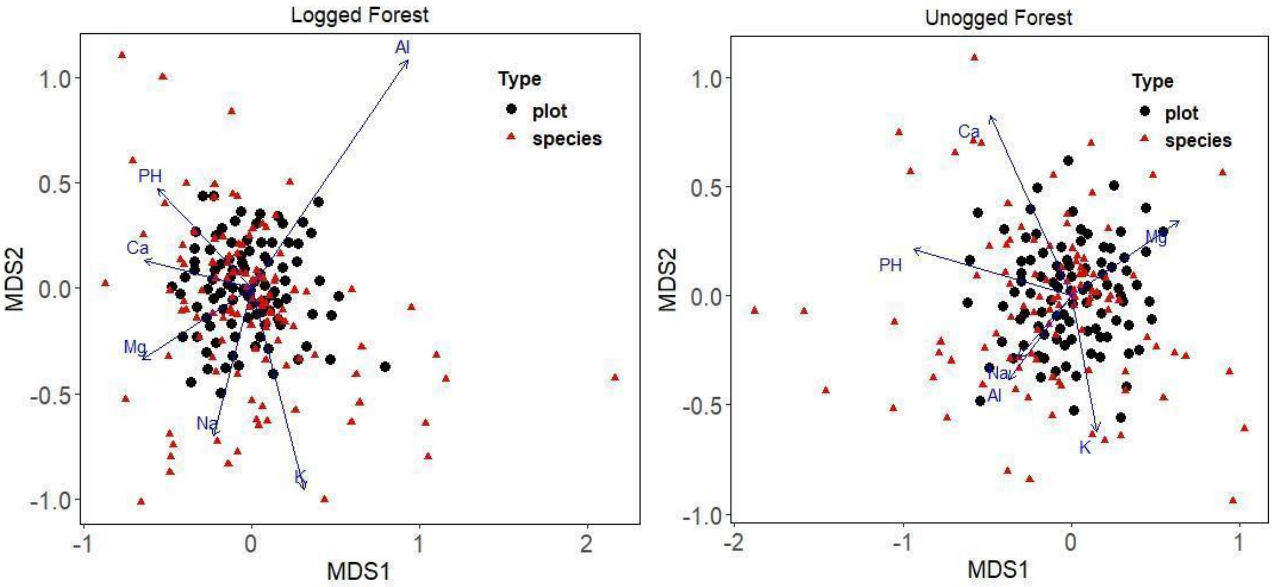

Figure 6: Non-metric multidimensional Scaling (NMDS) ordinations for Logged (left) and Unlogged (right) forests. Species abundances from both the canopy and regeneration layers were combined for each forest type to compute ordinations for species and plots. Mean concentrations of soil variables for each plot were fitted to the ordination to determine the influence of soil nutrients on the spatial distribution of species composition (plot) and individual species. The orientation of the arrow signifies the direction of the influence, while the length indicates the magnitude of the gradient or change.

Soil variable differences between the logged and unlogged forests were pronounced. CEC, Ca, K, Mg, and Na concentrations were higher in the unlogged forest, while soil pH was comparable between the two forest types (Table 5). This variation is also reflected in the NMDS ordination plots, where species distributions in the logged forest are more dispersed, suggesting greater heterogeneity in species composition compared to the unlogged forest (Fig. 5). Figure 5 illustrates the influence of soil variables on species' spatial distribution and composition (i.e., the similarity in species composition across plots). The stress values of 0.286 for the logged forest and 0.294 for the unlogged forest indicate a reasonable fit, suggesting that the visualized relationships between soil variables and species distributions are reliable representations of the data. The length of the arrows in the ordination plot further differentiates the primary soil nutrients influencing species distribution, with longer arrows representing stronger effects. In the logged forest, Al displayed the longest arrow, indicating that it has the strongest influence on species distribution. This suggests that Al plays a dominant role in shaping species composition in this forest type. Meanwhile, pH and K show opposing orientations, suggesting that plots with higher pH values are associated with lower K levels. The length of these arrows indicates that pH and K are also important factors but similar in strength to Mg, Na and Ca, though Al has a more pronounced influence.

In the unlogged forest, the ordination plot revealed a similar opposing relationship between Ca and K (Figure 5). Ca and K arrows point in nearly opposite directions, indicating that species in plots with

higher Ca levels tend to be associated with lower K levels, and vice versa. Additionally, the arrows for pH and Mg point in different directions from K and Ca, suggesting that these variables influence species distribution along separate gradients. The longer arrows for pH, K, Ca, and Mg highlight their major roles in shaping species composition in the unlogged forest. This pattern suggests that species composition in the unlogged forest is influenced by a broader range of soil variables acting along distinct gradients, whereas in the logged forest, Al plays a more dominant role.

Table 5: Mean (+ SE) soil ECEC (mmol), nutrient concentrations (mg/kg) and pH values for the logged and unlogged compartments. The mean differences are statistically significant for different letters between the logged and unlogged forests.

Compartment	CEC	Al	Ca	K	Mg	Na	pH
Logged	102.7 ± 4.3[b]	2.8 ± 0.9[a]	1672.2 ± 81.0[b]	113.1 ± 4.3[b]	169.2 ± 5.4[b]	4.7 ± 0.5[b]	6.2 ± 0.1[a]
Unlogged	146.2 ± 7.0[a]	5.1 ± 3.0[a]	2339.8 ± 133.7[a]	142.4 ± 6.5[a]	232.2 ± 8.4[a]	109.7 ± 8.5[a]	6.2 ± 0.1[a]

[1]CEC is the total cation exchange capacity in millimoles per 100 grams of soil (mmol/100g).

CHAPTER FIVE

5.0 Discussion

5.1 Tree species composition, diversity and recovery between logged and unlogged forests

To ascertain the influence of logging, tree species diversity is considered at the regeneration and canopy layers between the logged and unlogged forest. The results indicate that, in both the canopy and regeneration layers, the observed and extrapolated species diversity and abundance in the logged forest are higher and expected to increase comparatively. This implies that the gamma diversity is relatively higher in the logged forest (Figure 1 & 4). Interestingly, in the regeneration layer, although the logged forest recorded the highest species richness, the unlogged forest recorded species abundance which was 20% higher. The difference was rather marginal at the canopy layer. This suggests that there is to some extent a certain level of dominance in the unlogged forest relatively. This is indicated by the diversity indices; Pielou's evenness and Shannon diversity indices were higher in the logged forest than in the unlogged forest with this difference being statistically significant at both the canopy and regeneration layers, portraying relatively high evenness and species diversity in the logged forest. The difference in Inverse Simpson was not statistically significant for the regeneration layer but contrary at the canopy layer. It is believed that the sensitivity of Simpson and Inverse Simpson indices to rare species and its resulting variance contributed to the contrast in results. These results are in direct corroboration with the assertions of Connell, (2022), Gaui et al., (2019), Imai et al., (2012), Plumptre, (1996), Paul et al., (2004) and Shima et al., (2018) that the diversity status is highly influenced by the coexistence of early and late successional species dictated by the physiognomic stages of forest development. There is a rapid increase in species diversity from the initiation stage after the disturbance to a level where culmination is achieved and further establishment of dominant ones at the stem exclusion stage to depict the final species diversity of the forest to the old growth stage while there is the understory reinitiating awaiting disturbance to capitalize on canopy gap. Kosugi et al., (2016) estimated a minimum of 60 years to reach this level in tropical ecosystems. The intervention and interruption of forest resources including light and water patterns from the introduction of forest gaps promote this coexistence which elevates the species diversity. This agrees with the findings of Do et al., (2016), Ravnjak et al., (2022) and Win et al., (2012), that species diversity increased after disturbances in the forest canopy. After 70 years of logging, the findings suggest that the forest is within the transitional stage of stem exclusion and therefore can be regarded to be in the

matured stage and has recovered to a larger extent. This is reflected in the recovery variables assessed in this study. The differences in the canopy openness were not statistically significant (Okimat et al., 2023). The differences in basal area and number of trees per hectare were not statistically significant (Table 2). The differences observed in the aboveground biomass (AGB) may be attributed to the highly accumulated biomass by overgrown tree species with higher diameter classes of >100cm in the unlogged forest which was mainly *Cynometra alexandri*. The majority of the species reported in the IVI were also similar. The ecological guild of species in the different layers between the logged and unlogged foreets were similar (Figure 4). Therefore, these results are comparable to the outcomes of Bonnell et al., (2011) and Kosugi et al., (2016), that tropical forests require over 60 years to assume recovery to pre-logged conditions.

The differences in alpha diversities were not significant for the regeneration layer but were significant at the canopy layer with the logged forest recording the highest (Table 2). A similar trend was depicted at the forest level after standardizing the canopy and regeneration alpha diversity results; the canopy layer was more diverse than the regeneration layer (Figure 2). With the concept of succession already highlighted, it would be expected that the conditions that stimulated the establishment and coexistence of early and late successional species have resulted in what was once a cohort understory to successfully grow into the canopy of the remnant forest. These dynamics restrict the further recruitment of early successional species and result in a decreased richness in the regeneration layer as illustrated in Figure 4. In contrast, the coexistence of pioneer, intermediate and climax species in the canopy layer increases the species diversity (Berry et al., 2010; da Luz & Ferreira-Nunes, 2013; Imai et al., 2012, 2019; Molino & Sabatier, 2001). The findings from this study suggest that the pronounced variability from logging is prominent in the canopy layer after this long-term disturbance. Natural regeneration in selectively logged forests is also impeded for certain species requiring large diameter and low neighbourhood crowding to stimulate fruit production like in the case of *Khaya anthotheca* (Okimat et al., 2024) after the first rotation. It can be inferred that the current regeneration layer is a new cohort and species that were unable to grow into the canopy as explicated by Paul et al., (2004) as arrested succession at the undergrowth where there is expected high-level inter and intraspecific competition.

5.2 Similarity and dissimilarity of species composition

The following inferences are drawn from the ordination of species composition between the logged and unlogged forest; At the forest type (compartment) level, the logged forest encompasses the species composition of the unlogged with numerous unique dissimilar plots. At the forest layer level, the logged canopy layer exhibited a pronounced distinction with all other layers similar in species composition to the regeneration layer of the logged forest. Identical to the logged forest, the unlogged forest also possesses species composition at the regeneration layer highly dissimilar from the canopy layer with few plots overlapping. Regarding the individual plots within the different classes, the regeneration and canopy layers of the unlogged forest were highly similar, followed by the logged forest's canopy and regeneration layers (Figure 5). These results clearly indicate higher beta diversity in the logged forest and extremely higher uniqueness in species composition. It can be inferred that after the disturbance from selective logging, the species composition before logging is maintained or recovered in addition to other areas portraying a shift in species composition as portrayed by the canopy layer of the logged forest. A similar observation of distortion of the established spatial pattern of species composition after forest disturbance has been reported by Aoyagi et al., (2017). The execution of selective logging results in several patches in the forest which stimulates the germination and establishment of new species. Patches of low species richness due to dominance from pioneer or residual damage and patches with coexisting species of different ecological guilds are heterogeneously distributed. In this regard, the probability of encountering similar plots will be low and the resulting high dissimilarity. This is demonstrated in the findings of Hall et al., (2003) and Menge et al., (2023) through the concept of landscape variability and microclimate heterogeneity resulting in a mosaic of forest patches. This implies that similar species composition is expected for the unlogged forest due to homogenous conditions throughout the landscape and therefore high similarity in plot compositions as observed in the study. A similar observation of decreases in compositional similarity with increasing in-between distances of plots has been reported by Berry et al., (2008) and Imai et al., (2019). In addition, the species composition in the regeneration layer of the unlogged forest did not reflect in the canopy layer of the logged forest, though the results depict the old cohort of regeneration layer to have occupied the gap. It is expected that logging stimulates and enhances the establishment of new species and the remnant species after selectively logging the forest should be similar to the canopy layer of the unlogged forest than the regeneration layer. Furthermore, long-term canopy closure encourages high mortality of light demanders

and restructuring of the forest regeneration layer more towards recruitment and dominance of shade-tolerant species (Berry et al., 2008)The logged forest covers a substantial array of species composition. This is true because the logged forest encompasses different successional stages, resulting in higher spatial variability. After years of full recovery, as expressed in the development stage, the species composition of the logged is expected to be similar to that of the unlogged. However, this was partially true, as rather a large array of species composition was observed. The regeneration layer constitutes a wide spectrum of dissimilarity relative to the canopy layer of the logged forest. At the stand exclusion stage, a dense canopy is observed which hinders new saplings from growing to this level. The canopy continues to have one cohort (which in this forest is the remnant and the old cohort that filled the gaps from logging). The understory is also coerced to maintain a specific growth level at the reinitiating phase where seedlings, sampling and young trees beneath the canopy layer constitute multiple cohorts (Kosugi et al., 2016).

5.3 Influence of soil properties on the spatial distribution and composition of tree species

Soil macronutrients, especially N, P and K, as well as soil pH, have been shown to significantly influence tree growth and species distribution (John et al., 2007; Okimat et al., 2023; Manu et al 2022; Umana et al., 2021). This study demonstrates that a broader range of soil variables, such as Ca, Mg and Na also play significant roles, particularly in shaping species composition along distinct ecological gradients. In both forest types, these nutrients demonstrated similar effects to K and pH, with Al having the strongest influence on tree species distribution in the logged forest. This suggests that soil variables interact in complex ways, with different variables influencing community structure under varying forest conditions. These results corroborate with the findings of John et al., (2007) and Okimat et al. (2023), which emphasized the importance of pH and multiple soil variables like Ca, Mg, and Al in driving forest community structures similar to the primary nutrients.

The NMDS ordination plots underscored the significant role of Al in the logged forest, where its influence on species distribution was more pronounced than in the unlogged forest. This suggests that logging-induced changes in the soil environment may have intensified Al's impact on species composition. This aligns with studies by Kochian et al. (2004), which indicate that Al concentrations above 2 mg/kg can become toxic to plants, inhibiting root growth and regeneration, especially in well-drained soils like Lixisols. The higher influence of Al on

species distribution in the logged forest, despite its lower concentrations compared to the unlogged forest, may be attributed to the effect of logging on soil properties. Logging has been shown to alter soil structure and nutrient availability through physical disturbances such as soil compaction and reduced organic matter input (DeArmond 2019), which can result in lower concentrations of soil nutrients (Table 5) and significantly impact water infiltration (Sun et al., 2018). This together may have increased the mobility and bioavailability of Al in the logged forest, making it more accessible to plant roots (Ofoe et al., 2023; Zou, 2011; Wei et al., 2021). In contrast, the unlogged forest's higher CEC suggests the high concentrations of Ca, Mg, K and Na play a buffering role in mitigating the toxic effects of Al by competing for uptake at root sites and significantly promoting root growth, despite Al presence. Indeed, Ca and Mg have been shown to reduce Al toxicity by stabilizing plant cell walls and improving nutrient balance in soils (Rahman, 2022).

CHAPTER SIX

6.0 Conclusions and Recommendations

This study aims to assess the long-term impact of selective logging on tree species diversity and recovery in the Budongo Central Forest Reserve, Uganda. In this repute, species diversity and relative recovery are assessed after a portion of the reserve is subjected to logging with the primary unlogged forest as the control. The dissimilarity in species composition and the effect of logging on selected soil variables and their influence relative to soil pH and primary nutrients is also explored. The findings of this study indicated higher species diversity in the logged forest relative to the unlogged forest and revealed that the forest biomass and the dominating canopy trees recover to some extent after long-term disturbance similar to the unlogged forest after long-term recovery. This was achieved also because compartment N3 (logged forest) has been subjected to supervision after the logging and therefore benefitted from the passive management and the required rotation to encourage the coexistence of species of different ecological guilds and succession. To some extent, the economic value of the forest has been recovered since the main commercially valuable species are represented in the important dominating species after these years of forest recovery. This is crucial for decision-makers and conservationists regarding the management and restoration of disturbed tropical forests. Based on the results from this study, a minimum of 70 years is required for tropical forests to assume a certain level of recovery after disturbance, however, it is also prudent to acknowledge that this is also dependent on the disturbance intensity and the post-logging intervention. This addresses the practices of selective logging, it can be beneficial in realizing multiple objectives of tropical forests only if carefully undertaken and with its evolution to reduced impact logging in the tropics, it is recommended that compromises should not be tolerated in enforcing its strict adherence. The study has also revealed the expected physiognomic stage of a selectively logged forest after a certain period which is crucial in informing decisions regarding rotation, minimum harvestable diameter and cutting cycles.

High-level dissimilarity was observed between the regeneration and canopy layer of the logged forests. The unlogged forest (both canopy and regeneration) was similar in species composition to the regeneration layer of the logged forest. The dissimilarity was not completely exclusive as the overlap was more pronounced in the regeneration layer of the unlogged forest. Therefore, a certain level of shift in species composition is represented after logging, portrayed in the uniqueness of the logged canopy layer, however, this did not happen at the expense of the original species composition available in the study area.

A significant difference was observed for most of the soil variables between the logged and unlogged forest, however, a similar trend was depicted for their influence on the spatial distribution and composition of species except for Al and Mg. Numerous studies have highlighted the crucial role of soil pH and primary nutrients (K). Ca, Mg, Al, and Na also reflect this strength, and therefore, relevance is expected similarly in soil tree species interaction studies. This study assessed the relative influence based on the magnitude and orientation of the soil variables to the ordination axes to infer the relative importance of the soil variables. However, this is limited in ascertaining the response of each soil variable to the spatial distribution and composition of species, for example, the rank order importance. This should be considered in future studies. A crucial variable to be considered in the post-logging experiment is the bulk density of the soil which was not captured in this study, therefore research in the future should consider this and the extent to which it influences the nutrients after logging.

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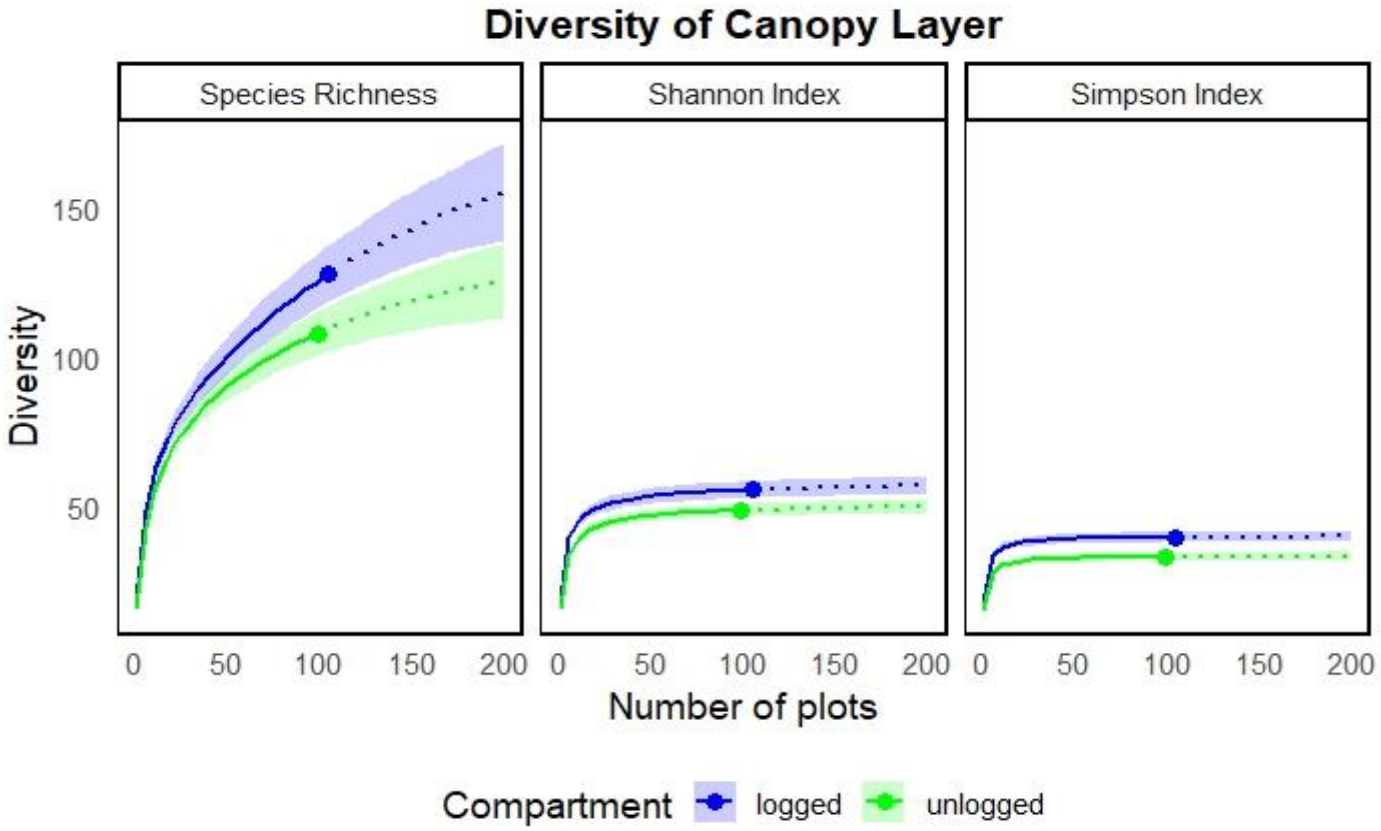
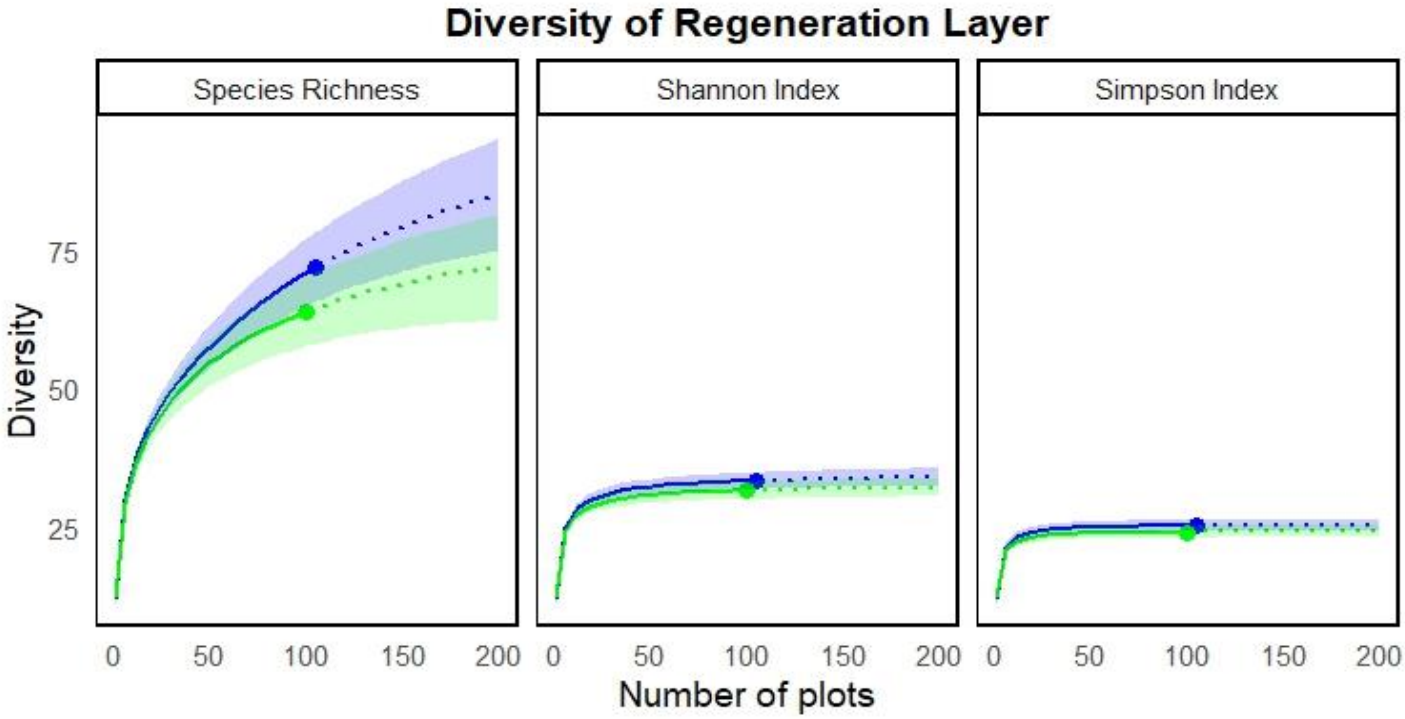
Appendices

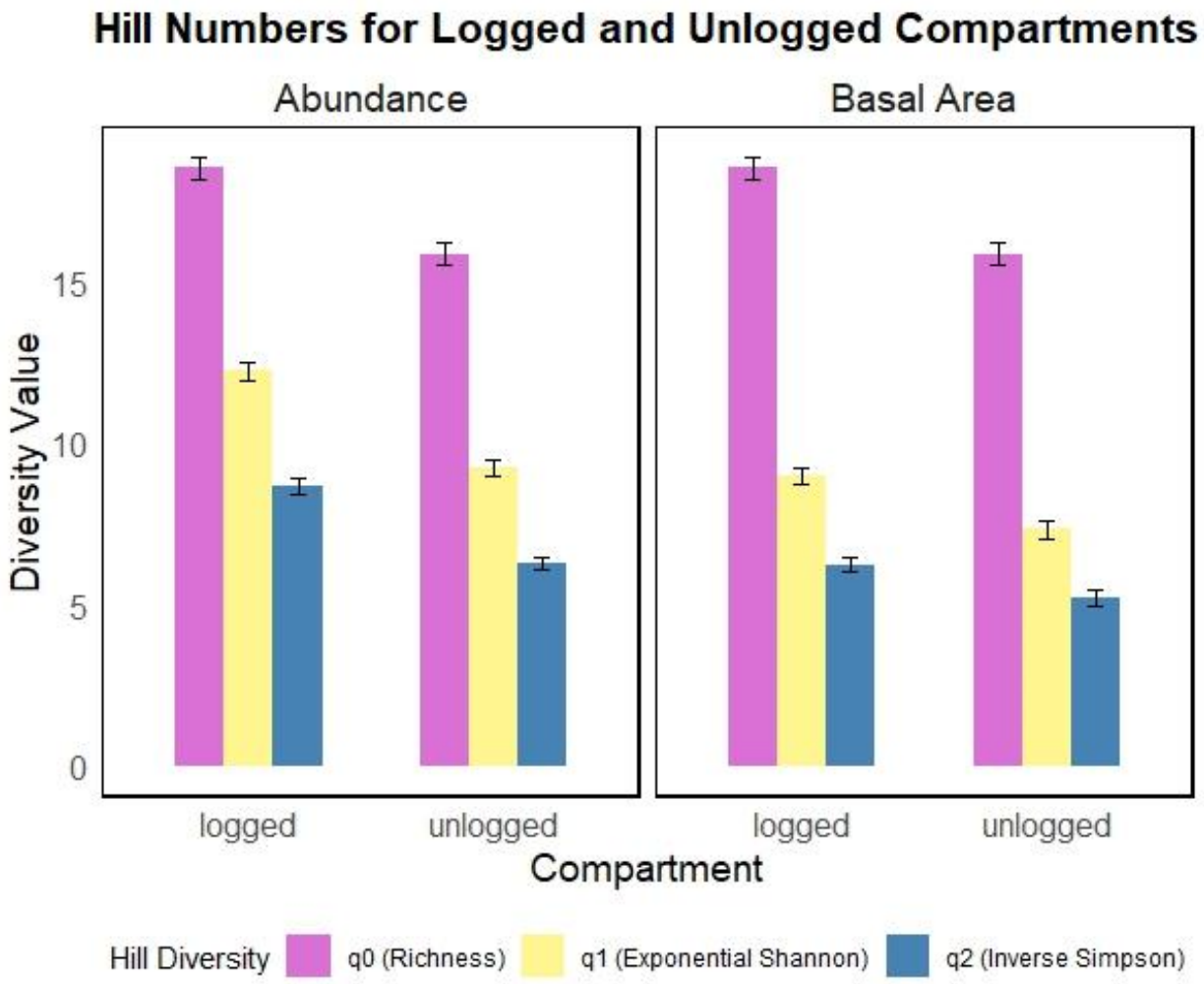
Appendix 1: Diversity indices

Appendix 1.1: Diversity estimation from Hill numbers between logged and unlogged forest





Appendix 1.2: Comparison of diversity indices estimated from basal area proportion and abundance between the forest types. Estimates were made for each plot before representing the mean plot values. Error bars represent standard error (SE).



APPENDIX 2: Tree species' ecological characteristics and functional traits for both canopy and regeneration layers from the sample plots at the Budongo Forest Reserve, Uganda. Tree species of DBH ≥ 10cm are categorized as canopy species and species of DBH < 10cm as regeneration layer. From the study sites, the logged forest is compartment N3 and the unlogged is compartment N15. Species that could not be identified are referred as **unknown** and therefore **unclassified** for their characteristics. NPLD: Non-pioneer Light demanders. The values recorded are the species abundance and areas represented as **0** indicate the absence of the species.

Genus	Species	Family	Code	Unlogged Canopy	Logged Canopy	Unlogged Regeneration	Logged Regeneration	Light guild	Wood density (g/cm3)
Aningeria	*altissima*	Sapotaceae	AAL	2	16	0	2	NPLD	0.6
Alstonia	*boonei*	Apocynaceae	AB	64	52	3	0	Pioneer	0.33
Albizia	*coriaria*	Mimosoideae	AC	0	1	0	20	Pioneer	0.52
Albizia	*glaberrima*	Mimosoideae	AGL	4	5	0	0	Pioneer	0.63
Alchornea	*laxiflora*	Euphorbiaceae	AL	8	58	11	117	Shade bearer	0.41
Albizia	*amara*	Mimosoideae	ALB	0	3	0	3	Pioneer	0.52
Allophyllus	*dummeri*	Sapindaceae	ALD	1	2	0	0	Shade bearer	0.45
Aneilema	*beninense*	Commelinaceae	AN	0	3	0	0	Shade bearer	0.45
Anchomanes	*difformis*	Araceae	ANC	0	1	0	0	Shade bearer	0.45
Antidesma	*spp*	Euphorbiaceae	ANI	0	1	0	0	Shade bearer	0.45
Antidesma	*laciniatum*	Euphorbiaceae	ANL	18	23	41	18	Shade bearer	0.63
Antiaris	*toxicaria*	Moraceae	ANT	15	100	30	66	Shade bearer	0.38
Aphania	*senegalensis*	Sapindaceae	APS	1	0	0	0	NPLD	0.76
Albizia	*zygia*	Mimosoideae	AZ	2	12	10	229	NPLD	0.55
Belonophora	*glomerata*	Rubiaceae	BEH	43	47	67	64	NPLD	0.72
Bequaertiodendron	*oblanceolatum*	Sapotaceae	BEO	13	1	31	0	Shade bearer	0.62
Blighia	*unijugata*	Sapindaceae	BLU	1	8	0	2	Shade bearer	0.55
Bosqueia	*phoberos*	Moraceae	BOS	0	1	0	0	Shade bearer	0.5
Broussonetia	*papyrifera*	Moraceae	BP	77	55	6	23	Pioneer	0.3
Baphia	*wollastonii*	Papilionoideae	BPW	1	11	0	0	Shade bearer	0.72
Bridelia	*brideliifolia*	Euphorbiaceae	BRB	15	2	0	0	Pioneer	0.77
Bridelia	*spp.*	Euphorbiaceae	BRD	1	0	0	0	Pioneer	0.55
Bridelia	*micrantha*	Euphorbiaceae	BRM	7	0	0	0	Pioneer	0.61
Chrysophyllum	*albidum*	Sapotaceae	CAL	34	43	145	76	Shade bearer	0.64
Coffea	*canephora*	Rubiaceae	CCA	0	6	0	0	Shade bearer	0.53
Celtis	*durandii*	Ulmaceae	CDU	3	137	0	6	NPLD	0.58
Clausena	*anisata*	Rutaceae	CLA	0	2	0	1	Shade bearer	0.55

Caloncoba	*schweinfurthii*	Flacourtaceae	CLS	18	68	1	7	Pioneer	0.57
Croton	*macrostachys*	Euphorbiaceae	CMC	3	9	1	0	Pioneer	0.75
Croton	*megalocarpus*	Euphorbiaceae	CMG	13	16	0	0	Pioneer	0.56
Celtis	*mildbraedii*	Ulmaceae	CMI	1113	887	89	205	Shade bearer	0.68
Chrysophyllum	*muerense*	Sapotaceae	CMU	4	6	0	7	Shade bearer	0.53
Canarium	*schweinfurthii*	Burseraceae	CNS	4	4	0	0	Pioneer	0.57
Cola	*gigantea*	Sterculiaceae	COG	8	6	70	32	NPLD	0.46
Cordia	*millenii*	Boraginaceae	COM	8	28	0	0	Pioneer	0.38
Cordia	*africana*	Boraginaceae	COR	0	1	0	0	Pioneer	0.41
Costus	*dubius*	Zingiberaceae	COS	33	13	1	3	Shade bearer	0.43
Cleistopholis	*patens*	Annonaceae	CP	10	8	3	1	Pioneer	0.24
Chrysophyllum	*perpulchrum*	Sapotaceae	CPR	46	52	8	27	Shade bearer	0.67
Cleistanthus	*polystachyus*	Euphorbiaceae	CROTON	0	1	0	0	Pioneer	0.76
Cleistanthus	*polystachyus*	Euphorbiaceae	CRP	27	1	0	0	Pioneer	0.5
Croton	*sylvaticus*	Euphorbiaceae	CSY	0	75	0	3	Pioneer	0.54
Celtis	*wightii*	Ulmaceae	CWI	38	33	15	12	Shade bearer	0.68
Celtis	*SPP.*	Ulmaceae	CWT	0	1	0	0	Shade bearer	0.55
Cynometra	*alexandri*	Caesalpinioideae	CYA	251	300	866	386	Shade bearer	0.71
Cyphomandra	*betaceae*	Solanaceae	CYS	0	1	0	0	Shade bearer	0.71
Celtis	*zenkeri*	Ulmaceae	CZE	130	309	65	2	NPLD	0.64
Dichrostachys	*cinerea*	Mimosoideae	DCC	0	4	0	0	Pioneer	0.69
Desplatsia	*dewevrei*	Tiliaceae	DD	11	24	1	5	Shade bearer	0.49
Diospyros	*abyssinica*	Ebenaceae	DIA	2	0	2	0	Pioneer	0.76
Drypetes	*spp.*	Euphorbiaceae	DPS	1	0	0	0	Pioneer	0.55
Drypetes	*spp.*	Euphorbiaceae	DS	17	2	10	2	Pioneer	0.48
Desplatsia	*spp.*	Tiliaceae	DSP	9	1	0	0	Shade bearer	0.48
Drypetes	*ugandensis*	Euphorbiaceae	DU	125	7	129	3	NPLD	0.74
Erythrina	*abyssinica*	Papilionoideae	EA	1	0	0	0	Pioneer	0.55
Ehretia	*cymosa*	Boraginaceae	EC	13	23	0	2	Pioneer	0.48
Erythrina	*excelsa*	Papilionoideae	EE	3	10	0	0	Pioneer	0.35
Entandrophragama	*angolense*	Meliaceae	ENA	13	13	8	1	NPLD	0.59
Entandrophragama	*cylindricum*	Meliaceae	ENC	4	1	2	2	NPLD	0.59
Entandrophragama	*utile*	Meliaceae	ENU	6	0	1	0	NPLD	0.59
Entandrophragama	*spp.*	Meliaceae	ENT	0	2	0	0	NPLD	0.65
Erythrophleum	*suaveolens*	Caesalpinioideae	ES	12	2	0	0	NPLD	0.7
Fagaropsis	*angolensis*	Rutaceae	FAA	1	1	0	0	NPLD	0.68

Fagara	*angolensis*	Rutaceae	FAN	7	4	0	5	NPLD	0.51
Ficus	*exasperata*	Moraceae	FE	1	16	0	0	Pioneer	0.36
Ficus	*lingua*	Moraceae	FL	2	0	0	0	Pioneer	0.38
Fagaropsis	*angolensis*	Rutaceae	FLE	4	0	0	0	NPLD	0.68
Ficus	*mucuso*	Moraceae	FM	0	2	0	0	Pioneer	0.4
Fagara	*macrophylla*	Rutaceae	FMA	0	1	0	0	Shade bearer	0.62
Fagara	*mildbraedii*	Rutaceae	FMU	0	5	0	0	Shade bearer	0.4
Ficus	*natalensis*	Moraceae	FN	1	0	0	0	Pioneer	0.38
Ficus	*spp.*	Moraceae	FNA	0	1	0	0	Pioneer	0.38
Ficus	*ottoniaefolia*	Moraceae	FO	1	0	0	0	Pioneer	0.38
Ficus	*vogelana*	Moraceae	FOV	1	0	0	0	Pioneer	0.38
Ficus	*polita*	Moraceae	FPO	0	1	0	0	Pioneer	0.39
Ficus	*saussureana*	Moraceae	FSS	1	3	0	0	Pioneer	0.4
Ficus	*stipulifera*	Moraceae	FST	0	1	0	0	Pioneer	0.35
Ficus	*sur*	Moraceae	FSU	9	40	0	2	Pioneer	0.37
Funtumia	*africana*	Apocynaceae	FUA	23	0	2	0	Shade bearer	0.4
Funtumia	*elastica*	Apocynaceae	FUE	285	952	30	75	NPLD	0.48
Ficus	*variifolia*	Moraceae	FVR	4	23	2	0	Pioneer	0.44
Guarea	*cedrata*	Meliaceae	GC	26	31	17	11	Shade bearer	0.57
Guarea	*spp.*	Meliaceae	GH	0	1	0	0	Shade bearer	0.55
Glyphaea	*spp.*	Tiliaceae	GM	0	1	0	0	Shade bearer	0.48
Glyphaea	*brevis*	Tiliaceae	GR	0	4	0	0	Shade bearer	0.63
Greenwayodendron	*suaveolens*	Annonaceae	GS	64	63	84	72	Shade bearer	0.66
Holoptelea	*grandis*	Ulmaceae	HG	13	42	3	8	Pioneer	0.54
Irvingia	*gabonensis*	Ixonanthes	IG	1	0	0	0	Shade bearer	0.71
Khaya	*anthotheca*	Meliaceae	KA	75	123	109	145	NPLD	0.53
Klainedoxa	*gabonensis*	Balanitaceae	KLG	7	17	1	0	NPLD	0.72
Lannea	*welwitchii*	Anacardiaceae	LA	0	1	0	0	Pioneer	0.55
Lasiodiscus	*mildbreadii*	Rhamnaceae	LM	1160	492	2957	1848	Shade bearer	0.8
Lovoa	*trichilioides*	Meliaceae	LOVAO	1	2	0	1	NPLD	0.61
Leptaulus	*daphnoides*	Icacinaceae	LPD	6	1	1	0	Shade bearer	0.49
Leptonychia	*mildbraedii*	Sterculiaceae	LPM	20	22	4	18	Shade bearer	0.57
Lovoa	*swynnertonii*	Meliaceae	LSY	3	0	0	0	NPLD	0.5
Lovoa	*trichilioides*	Meliaceae	LT	0	1	0	0	NPLD	0.61
Lychnodiscus	*cerospermus*	Sapindaceae	LYC	35	24	30	9	Shade bearer	0.71
Mammea	*africana*	Guttiferae	MA	16	22	65	66	Shade bearer	0.62

Mimusops	*bagshawei*	Sapotaceae	MBG	2	0	0	0	Shade bearer	0.65
Maerua	*duchesnei*	Capparaceae	MD	1	1	2	1	Shade bearer	0.53
Margaritaria	*discoidea*	Euphorbiaceae	MDI	16	68	7	10	Pioneer	0.56
Maesopsis	*eminii*	Rhamnaceae	ME	27	49	0	1	Pioneer	0.43
Maesa	*lanceolata*	Myrsinaceae	MEL	9	20	0	11	Shade bearer	0.41
Milicia	*excelsa*	Moraceae	MEX	5	21	0	0	NPLD	0.41
Mildbraediodendron	*excelsum*	Caesalpinioideae	MIE	4	1	6	1	NPLD	0.83
Monodora	*myristica*	Annonaceae	MIL	2	0	0	0	Shade bearer	0.83
Macaranga	*barteri*	Euphorbiaceae	ML	1	0	0	0	Pioneer	0.4
Mallotus	*oppositifolias*	Euphorbiaceae	MLS	0	1	0	0	NPLD	0.4
Macaranga	*monandra*	Euphorbiaceae	MM	3	0	0	0	Pioneer	0.48
Manilkara	*obovata*	Sapotaceae	MO	0	2	1	0	Shade bearer	0.48
Monodora	*angolensis*	Annonaceae	MOA	0	21	1	15	Shade bearer	0.51
Morus	*lactea*	Moraceae	MOL	8	8	1	2	Shade bearer	0.58
Markhamia	*platycalyx*	Bignoniaceae	MP	1	0	0	0	Pioneer	0.45
Manilkara	*butugi*	Sapotaceae	MRB	1	0	0	0	Shade bearer	0.5
Macaranga	*schweinfurtii*	Euphorbiaceae	MS	6	0	0	0	Pioneer	0.4
Mitragyna	*stipulosa*	Rubiaceae	MST	5	6	0	0	NPLD	0.52
Myrianthus	*holstii*	Moraceae	MYH	39	96	37	26	Shade bearer	0.54
Newtonia	*buchananii*	Fabaceae	NUE	1	0	0	0	Pioneer	0.26
Geophila	*obvallata*	Rubiaceae	OB	3	0	4	0	Shade bearer	0.65
Ouratea	*hiernii*	Ochnaceae	OUH	11	3	5	11	Pioneer	0.5
Pancovia	*turbinata*	Sapindaceae	PAT	0	1	0	0	Shade bearer	0.45
Pyrencantha	*sylvestris*	Icacinaceae	PD	0	1	0	0	Shade bearer	0.48
Paropsia	*guineensis*	Passifloraceae	PG	0	1	0	0	Shade bearer	0.45
Margaritaria	*discoidea*	Euphorbiaceae	PHD	0	1	0	0	Shade bearer	0.6
Picralima	*nitida*	Apocynaceae	PN	0	2	0	0	Shade bearer	0.55
Psidium	*guajava*	Myrtaceae	PSG	0	6	0	10	Pioneer	0.45
Pseudospondias	*microcarpa*	Anacardiaceae	PSM	4	4	0	0	Shade bearer	0.5
Pycnanthus	*angolensis*	Myristicaceae	PYA	0	2	0	0	NPLD	0.54
Ricinodendron	*heudelotii*	Euphorbiaceae	RH	3	15	0	0	Pioneer	0.3
Rinorea	*ardisiaeflora*	Violaceae	RI	634	191	2220	1011	Shade bearer	0.65
Unknown	*Unknown*	unclassified	UNK7	0	1	0	0	Unclassified	0.65
Schrebera	*arborea*	Oleaceae	SA	1	1	0	0	Pioneer	0.6
Spathodea	*campanulata*	Bignoniaceae	SC	2	0	0	0	Pioneer	0.25
Sapium	*ellipticum*	Euphorbiaceae	SE	2	0	0	1	Pioneer	0.5

Spondianthus	*preussii*	Euphorbiaceae	SPI	0	4	0	0	Pioneer	0.64
Strombosia	*scheffleri*	Olacaceae	SS	1	1	0	0	Shade bearer	0.7
Strychnos	*innocua*	Loganiaceae	STR	0	7	0	3	Shade bearer	0.6
Tapura	*fischeri*	Dichapetelaceae	TAF	64	126	160	108	Shade bearer	0.68
Tetrorchidium	*didymostemon*	Euphorbiaceae	TD	1	2	0	0	Pioneer	0.53
Tabernaemontana	*holstii*	Apocynaceae	TH	43	31	134	113	Shade bearer	0.55
Teclea	*nobilis*	Rutaceae	TN	5	10	32	89	Shade bearer	0.7
Trema	*orientalis*	Ulmaceae	TO	11	1	0	0	Pioneer	0.3
Trichilia	*dregeana*	Meliaceae	TRD	0	3	0	0	NPLD	0.6
Trichilia	*prieuriana*	Meliaceae	TRP	69	142	77	82	Shade bearer	0.63
Trichilia	*rubescens*	Meliaceae	TRR	298	245	74	54	NPLD	0.55
Tetrapleura	*tetraptera*	Mimosoideae	TT	23	15	0	7	Pioneer	0.62
Uvariopsis	*congensis*	Annonaceae	UC		20	5	50	Shade bearer	0.65
Unknown	*Unknown*	unclassified	UNK	6	25	17	42	Unclassified	0.65
Unknown	*Unknown*	unclassified	UNK4	0	1	0	0	Unclassified	0.65
Unknown	*Unknown*	unclassified	UNKI	0	0	1	0	Unclassified	0.65
Unknown	*Unknown*	unclassified	UNK2	0	0	4	2	Unclassified	0.65
Unknown	*Unknown*	unclassified	UNK3	0	1	2	0	Unclassified	0.65
Unknown	*Unknown*	unclassified	UNK5	0	1	0	0	Unclassified	0.65
Unknown	*Unknown*	unclassified	UNK6	0	2	0	0	Unclassified	0.65
Vangueria	*apiculata*	ubiaceae	VAM	0	5	0	0	Shade bearer	0.6
Citropsis	*articulata*	Rutaceae	CIT	0	0	8	4	Shade bearer	0.5
Cleistanthus	*polystachyus*	Euphorbiaceae	CTP	0	0	0	2	Shade bearer	0.6
Lasiodiscus	*spp.*	Rhamnaceae	LMM	0	0	1	1	Shade bearer	0.8
Manilkara	*dawei*	Sapotaceae	MRK	0	0	0	1	Shade bearer	0.55
Unknown	*Unknown*	unclassified	UNK7	0	0	2	2	Shade bearer	0.65
Raphia	*farinifera*	Agavaceae	RA	0	0	0	1	Pioneer	0.48
Renealmia	*spp.*	Zingiberaceae	R	0	0	1	0	Shade bearer	0.45
Rinorea	*spp.*	Violaceae	R1	0	0	3	0	Shade bearer	0.56
Rothmania	*spp.*	Rubiaceae	RD	0	0	64	9	Shade bearer	0.55
Treculia	*africana*	Moraceae	TR	0	0	0	1	Pioneer	0.4

Appendix 3.1: Soil variables for Logged Forest. Values represent the mean measurement for each plot for that soil variable.

Plot.id	Compartment	CEC	Al	Ca	K	Mg	Na	PH	Fe
N3_1_10	Logged	106.8	0.1	1726.1	184.7	173.5	6.2	6.5	<0.03
N3_1_20	Logged	146.8	0.1	2537.2	113.3	192.2	8.0	6.5	<0.03
N3_1_30	Logged	102.1	0.1	1738.0	85.4	145.2	2.8	6.4	<0.03
N3_1_40	Logged	42.6	0.4	677.6	54.8	55.4	1.8	5.7	<0.03
N3_1_50	Logged	113.5	0.1	1958.8	117.1	131.7	7.8	6.3	<0.03
N3_1_60	Logged	111.8	0.1	1863.6	159.7	155.5	1.8	6.3	<0.03
N3_1_70	Logged	120.8	0.1	1940.7	156.2	221.6	2.3	6.4	<0.03
N3_1_80	Logged	32.2	2.0	498.6	34.5	68.2	1.9	5.7	<0.03
N3_1_90	Logged	65.0	0.1	1049.2	85.6	90.7	6.5	5.9	<0.03
N3_1_100	Logged	44.0	25.5	565.6	96.1	80.0	3.0	5.2	<0.03
N3_1_110	Logged	142.0	0.1	2332.6	205.4	240.4	4.0	7.0	<0.03
N3_1_120	Logged	68.2	0.8	1086.0	76.2	101.6	0.9	5.7	<0.03
N3_1_130	Logged	134.2	0.1	2254.0	119.5	204.0	4.1	6.3	<0.03
N3_1_140	Logged	110.7	0.1	1838.1	107.7	174.8	4.2	6.3	<0.03
N3_1_150	Logged	139.4	0.1	2363.6	125.2	212.6	5.0	7.1	<0.03
N3_1_160	Logged	78.1	0.1	1169.2	108.7	153.2	5.0	6.0	<0.03
N3_1_170	Logged	251.0	0.1	4709.3	185.9	132.6	2.7	6.9	<0.03
N3_1_180	Logged	66.4	2.0	839.8	64.2	233.1	7.2	5.8	<0.03
N3_1_190	Logged	91.4	0.1	1361.9	57.5	241.3	16.9	6.4	<0.03
N3_1_200	Logged	95.9	0.3	1324.4	160.8	253.8	6.4	5.9	<0.03
N3_2_10	Logged	211.0	0.1	3886.6	176.5	148.2	6.3	7.6	<0.03
N3_2_20	Logged	103.5	0.1	1753.1	105.0	151.8	2.0	6.9	<0.03
N3_2_30	Logged	160.7	0.1	2924.4	130.7	136.3	0.5	7.4	<0.03
N3_2_40	Logged	196.6	0.1	3474.0	130.6	223.7	3.0	6.9	<0.03
N3_2_50	Logged	138.5	0.1	2289.9	172.0	221.8	1.2	6.8	<0.03
N3_2_60	Logged	92.8	0.1	1487.9	88.6	175.6	0.8	6.5	<0.03
N3_2_70	Logged	90.7	0.1	1532.4	88.0	127.5	1.0	6.5	<0.03
N3_2_80	Logged	123.0	0.1	2160.2	107.6	138.9	1.6	6.6	<0.03
N3_2_90	Logged	161.6	0.1	2780.0	173.9	212.6	3.8	6.8	<0.03
N3_2_100	Logged	189.3	0.1	3263.0	123.9	267.2	5.4	6.8	<0.03
N3_2_110	Logged	165.6	0.1	2853.8	153.3	206.7	4.5	6.5	<0.03
N3_2_120	Logged	134.5	0.1	2257.8	113.9	199.6	3.2	6.4	<0.03
N3_2_130	Logged	95.3	0.1	1566.0	74.1	155.9	2.8	6.4	<0.03
N3_2_140	Logged	189.0	0.1	3318.5	179.0	216.4	2.4	6.9	<0.03
N3_2_150	Logged	89.3	0.1	1494.5	98.6	132.0	2.1	6.6	<0.03
N3_2_160	Logged	85.3	0.1	1388.0	157.9	111.2	4.0	6.3	<0.03
N3_2_170	Logged	149.1	0.1	2533.6	186.0	192.0	1.9	6.5	<0.03
N3_2_180	Logged	91.7	0.1	1486.8	69.0	146.2	2.6	6.1	<0.03
N3_2_190	Logged	104.4	0.1	1665.2	101.1	186.7	4.2	6.1	<0.03
N3_2_200	Logged	97.5	0.1	1569.4	96.3	170.6	3.4	6.2	<0.03
N3_2_210	Logged	118.5	0.1	1963.2	134.4	189.8	2.5	6.5	<0.03

N3_2_220	Logged	95.0	0.1	1385.2	123.4	243.5	2.8	6.2	<0.03
N3_2_230	Logged	44.5	3.2	606.9	69.6	118.2	1.5	5.7	<0.03
N3_2_240	Logged	43.6	18.6	594.6	35.6	105.2	9.5	5.3	<0.03
N3_2_250	Logged	68.4	0.9	891.9	49.9	233.6	13.9	5.7	<0.03
N3_3_10	Logged	73.5	2.8	1004.2	82.0	215.5	3.7	5.7	<0.03
N3_3_20	Logged	65.8	27.0	935.8	82.0	130.6	8.4	5.4	<0.03
N3_3_30	Logged	88.9	0.1	1461.2	86.0	134.2	3.0	6.0	<0.03
N3_3_40	Logged	57.2	0.3	830.5	116.0	110.1	24.7	5.7	<0.03
N3_3_50	Logged	73.5	0.1	1116.4	117.7	164.5	1.3	6.4	<0.03
N3_3_60	Logged	74.0	0.1	1270.7	45.9	100.0	1.6	6.2	<0.03
N3_3_70	Logged	84.7	0.1	1557.0	11.3	73.1	0.2	6.6	<0.03
N3_3_80	Logged	70.1	0.1	1189.2	69.1	100.9	1.6	6.5	<0.03
N3_3_90	Logged	79.7	0.1	1211.4	102.8	165.8	1.4	6.1	<0.03
N3_3_110	Logged	65.9	0.1	1019.0	35.9	144.8	0.5	6.1	<0.03
N3_3_120	Logged	125.1	0.1	2097.3	177.9	179.2	4.7	6.6	<0.03
N3_3_130	Logged	95.4	0.1	1516.7	133.7	175.0	2.5	6.4	<0.03
N3_3_140	Logged	139.0	0.1	2341.5	144.4	203.7	6.5	6.4	<0.03
N3_3_150	Logged	129.6	0.1	2206.8	112.0	189.3	2.5	6.6	<0.03
N3_3_160	Logged	156.8	0.1	2711.6	174.6	193.5	3.0	6.5	<0.03
N3_3_170	Logged	133.3	0.1	2199.0	214.5	205.0	0.2	6.5	<0.03
N3_3_180	Logged	200.6	0.1	3472.8	109.4	278.1	0.7	6.5	<0.03
N3_3_190	Logged	141.8	0.1	2342.2	133.9	238.5	1.2	6.4	<0.03
N3_3_200	Logged	106.6	0.1	1648.4	114.4	244.5	0.2	6.4	<0.03
N3_4_10	Logged	100.7	0.1	1603.3	114.6	182.5	0.9	6.0	<0.03
N3_4_30	Logged	104.0	0.1	1754.2	108.4	155.5	0.5	6.4	<0.03
N3_4_40	Logged	66.2	8.1	996.9	40.0	135.0	0.5	5.5	<0.03
N3_4_50	Logged	62.5	2.2	1011.7	47.0	95.0	1.0	5.6	<0.03
N3_4_60	Logged	75.0	0.1	1149.6	100.3	154.0	0.5	6.0	<0.03
N3_4_70	Logged	82.6	0.1	1430.8	56.3	110.6	0.2	6.4	<0.03
N3_4_90	Logged	42.1	48.5	385.4	78.2	154.4	12.7	5.1	0.2
N3_4_100	Logged	73.9	18.2	920.4	72.2	251.1	21.4	5.2	2.0
N3_4_110	Logged	45.0	0.3	675.6	50.0	106.0	1.3	5.8	<0.03
N3_4_120	Logged	95.9	0.1	1560.7	125.2	164.0	2.0	6.5	<0.03
N3_4_130	Logged	107.8	0.1	1777.8	94.9	170.4	1.1	6.2	<0.03
N3_4_140	Logged	100.1	0.1	1571.3	170.1	190.1	0.6	6.4	<0.03
N3_4_150	Logged	91.6	0.1	1551.6	101.1	114.8	1.4	6.1	<0.03
N3_4_160	Logged	107.1	0.1	1769.4	116.0	157.2	1.7	6.1	<0.03
N3_4_170	Logged	172.8	0.1	2946.9	184.8	238.5	3.5	6.5	<0.03
N3_4_180	Logged	204.9	0.1	3399.2	194.6	349.3	1.8	6.7	<0.03
N3_4_190	Logged	181.4	0.1	3042.5	200.0	276.3	2.2	6.6	<0.03
N3_4_200	Logged	136.7	0.1	2144.3	186.8	290.6	0.5	6.6	<0.03
N3_4_210	Logged	112.7	0.1	1913.1	126.6	152.8	0.6	6.5	<0.03
N3_5_10	Logged	97.1	0.3	1554.2	120.8	145.7	8.2	5.9	<0.03
N3_5_20	Logged	90.3	0.1	1520.8	100.4	119.8	5.0	6.2	<0.03
N3_5_30	Logged	102.4	0.1	1711.3	124.7	140.6	9.2	6.3	<0.03
N3_5_40	Logged	137.8	0.1	2198.2	148.8	263.5	9.1	6.3	<0.03

N3_5_50	Logged	94.0	0.1	1484.5	106.5	180.7	9.6	6.1	<0.03
N3_5_60	Logged	103.7	0.1	1589.1	109.4	231.0	9.7	6.1	<0.03
N3_5_70	Logged	79.4	0.1	1128.3	125.3	201.1	10.2	5.9	<0.03
N3_5_80	Logged	59.6	20.1	818.5	139.4	117.2	10.0	5.4	<0.03
N3_5_90	Logged	69.0	0.1	1080.9	124.5	99.8	11.9	5.7	<0.03
N3_5_100	Logged	90.8	0.1	1441.3	120.4	162.3	9.8	6.0	<0.03
N3_5_110	Logged	79.4	0.1	1220.6	149.7	152.2	7.3	6.2	<0.03
N3_5_120	Logged	86.3	0.1	1378.8	122.8	143.2	9.4	6.1	<0.03
N3_5_130	Logged	74.4	0.1	1167.8	117.0	110.9	9.3	5.8	<0.03
N3_5_140	Logged	56.6	0.1	812.8	98.6	138.4	7.4	6.2	<0.03
N3_5_150	Logged	30.7	58.6	262.4	41.7	91.4	19.8	5.2	<0.03
N3_5_160	Logged	58.4	34.6	672.2	73.0	193.7	26.1	5.2	<0.03
N3_5_170	Logged	32.3	5.0	413.5	59.5	78.1	1.5	5.4	<0.03
N3_5_180	Logged	140.1	0.1	2360.8	141.7	217.5	3.5	6.5	<0.03
N3_5_190	Logged	61.7	0.4	888.2	101.8	132.9	3.6	5.6	<0.03
N3_5_200	Logged	53.6	0.1	782.1	67.2	127.7	0.2	6.0	<0.03
N3_5_210	Logged	57.0	0.1	722.2	104.2	177.2	3.6	5.7	<0.03

For Plot ID, N3 is the logged compartment and N15 is the unlogged compartment.

Appendix 3.2: Soil variables for Logged Forest. Values represent the mean measurement for each plot for that soil variable.

Plot.id	compartment	CEC	Al	Ca	K	Mg	Na	PH	Fe
N15_1_10	Unlogged	125.3	0.1	1974.4	185.4	226.7	6.5	6.2	<0.03
N15_1_20	Unlogged	100.1	0.1	1414.5	118.8	283.8	3.8	6.1	<0.03
N15_1_30	Unlogged	131.5	0.1	2034.1	168.0	264.5	8.6	6.0	<0.03
N15_1_40	Unlogged	54.6	47.6	508.3	113.1	173.0	8.1	5.2	<0.03
N15_1_50	Unlogged	71.0	0.1	983.2	61.0	215.7	4.2	5.8	<0.03
N15_1_60	Unlogged	36.8	0.1	495.0	27.6	115.4	25.9	6.1	<0.03
N15_1_70	Unlogged	49.4	2.8	730.0	55.4	101.4	5.0	5.4	<0.03
N15_1_80	Unlogged	55.5	1.1	814.0	46.8	114.7	5.8	5.5	<0.03
N15_1_90	Unlogged	150.5	0.1	2567.8	94.0	221.6	8.6	6.5	<0.03
N15_1_100	Unlogged	386.9	0.1	7361.9	277.3	145.5	8.2	7.8	<0.03
N15_1_110	Unlogged	225.5	0.1	4069.8	114.8	220.3	8.5	6.9	<0.03
N15_1_120	Unlogged	202.3	0.1	3447.2	165.3	294.0	9.4	6.7	<0.03
N15_1_130	Unlogged	306.7	0.1	5647.7	164.2	240.2	8.7	7.3	<0.03
N15_1_140	Unlogged	164.8	0.1	2669.0	197.2	309.7	6.2	6.9	<0.03
N15_1_150	Unlogged	90.7	0.1	1507.2	100.5	115.5	8.0	6.1	<0.03
N15_1_160	Unlogged	109.9	0.1	1724.1	140.5	217.1	9.2	6.3	<0.03
N15_1_170	Unlogged	98.7	0.1	881.9	31.9	127.2	4.6	5.8	<0.03
N15_1_180	Unlogged	29.3	2.6	424.9	33.8	59.4	3.6	5.7	<0.03
N15_1_190	Unlogged	32.3	2.7	469.7	45.1	70.9	5.2	5.6	<0.03
N15_1_200	Unlogged	116.9	0.1	1569.8	158.2	356.5	6.8	5.9	<0.03
N15_2_10	Unlogged	183.0	0.1	3208.5	143.2	218.5	5.6	6.7	<0.03
N15_2_20	Unlogged	138.2	0.3	2186.3	161.8	191.3	11.4	5.5	<0.03

N15_2_30	Unlogged	177.1	0.1	2902.2	137.8	328.8	6.6	6.6	<0.03
N15_2_40	Unlogged	60.4	1.9	830.6	71.0	135.7	3.9	5.5	<0.03
N15_2_50	Unlogged	118.5	0.1	1647.7	159.4	316.9	5.7	5.8	<0.03
N15_2_60	Unlogged	73.9	1.1	729.4	220.8	295.6	5.5	5.6	<0.03
N15_2_70	Unlogged	79.3	0.1	1131.1	88.8	203.1	4.8	5.5	<0.03
N15_2_80	Unlogged	43.8	8.9	528.1	127.7	114.8	3.8	5.1	<0.03
N15_2_90	Unlogged	121.2	0.1	1884.8	140.8	212.0	140.8	5.9	<0.03
N15_2_100	Unlogged	86.9	1.2	1409.2	80.0	133.2	80.0	5.4	<0.03
N15_2_110	Unlogged	96.1	0.1	1409.2	160.4	178.4	160.4	5.7	<0.03
N15_2_120	Unlogged	154.3	0.1	2542.0	202.4	163.6	202.4	6.4	<0.03
N15_2_130	Unlogged	132.1	0.1	2024.4	202.8	208.0	202.8	6.2	<0.03
N15_2_140	Unlogged	142.1	0.1	2175.6	160.4	272.4	160.4	6.3	<0.03
N15_2_150	Unlogged	125.9	0.1	2041.2	160.8	157.6	160.8	6.2	<0.03
N15_2_160	Unlogged	142.1	0.1	2433.2	120.4	150.8	120.4	6.7	<0.03
N15_2_170	Unlogged	296.6	0.1	5241.2	231.2	232.4	231.2	7.5	<0.03
N15_2_180	Unlogged	63.3	0.1	988.4	91.6	93.2	91.6	6.4	<0.03
N15_2_190	Unlogged	194.7	0.1	3200.4	138.0	309.6	138.0	6.6	<0.03
N15_2_200	Unlogged	84.0	0.1	1167.2	69.2	254.8	69.2	5.7	<0.03
N15_3_10	Unlogged	206.0	0.1	3425.2	217.6	243.6	217.6	6.7	<0.03
N15_3_20	Unlogged	189.9	0.1	2942.8	254.4	309.2	254.4	6.4	<0.03
N15_3_30	Unlogged	225.3	0.1	3598.8	248.0	347.6	248.0	6.5	<0.03
N15_3_40	Unlogged	187.5	0.1	2857.6	242.0	342.0	242.0	6.2	<0.03
N15_3_50	Unlogged	195.0	0.1	3034.0	224.8	341.2	224.8	6.8	<0.03
N15_3_60	Unlogged	128.5	0.1	1934.8	206.8	214.4	206.8	6.2	<0.03
N15_3_70	Unlogged	193.8	0.1	3330.4	140.8	217.2	140.8	6.6	<0.03
N15_3_80	Unlogged	231.2	0.1	4132.4	134.8	190.0	134.8	6.9	<0.03
N15_3_90	Unlogged	292.7	0.1	5028.8	155.6	376.4	155.6	7.2	<0.03
N15_3_100	Unlogged	176.0	0.1	2861.6	170.4	260.8	170.4	6.6	<0.03
N15_3_110	Unlogged	185.0	0.1	2940.8	205.2	292.4	205.2	6.3	<0.03
N15_3_120	Unlogged	160.3	0.1	2663.2	146.8	210.0	146.8	6.2	<0.03
N15_3_130	Unlogged	246.0	0.1	4215.2	152.0	305.2	152.0	6.6	<0.03
N15_3_140	Unlogged	98.9	0.1	1450.4	102.4	236.8	102.4	5.9	<0.03
N15_3_150	Unlogged	111.3	0.1	1673.6	98.8	255.2	98.8	5.6	<0.03
N15_3_160	Unlogged	151.7	0.1	2381.6	168.4	257.6	168.4	6.2	<0.03
N15_3_170	Unlogged	233.1	0.1	3979.2	183.2	266.4	183.2	6.3	<0.03
N15_3_180	Unlogged	206.1	0.1	3628.8	134.8	191.2	134.8	6.2	<0.03
N15_3_190	Unlogged	179.6	0.1	2854.4	125.6	346.0	125.6	5.8	<0.03
N15_3_200	Unlogged	187.9	0.1	2937.6	92.0	424.8	92.0	6.0	<0.03
N15_4_10	Unlogged	68.7	15.2	1026.8	64.8	132.4	64.8	5.0	<0.03
N15_4_30	Unlogged	231.4	0.1	3914.0	185.2	283.6	185.2	6.9	<0.03
N15_4_40	Unlogged	230.9	0.1	3929.2	138.4	306.8	138.4	6.8	<0.03
N15_4_50	Unlogged	145.7	0.1	2246.0	120.8	307.2	120.8	6.1	<0.03
N15_4_60	Unlogged	197.7	0.1	3474.4	115.6	198.4	115.6	6.6	<0.03
N15_4_70	Unlogged	129.5	0.1	1892.0	208.8	251.6	208.8	6.2	<0.03
N15_4_80	Unlogged	218.3	0.1	3441.2	318.8	298.0	318.8	6.8	<0.03
N15_4_90	Unlogged	151.1	0.1	2510.4	119.2	213.6	119.2	6.5	<0.03

N15_4_100	Unlogged	133.7	0.1	1903.2	223.6	283.2	223.6	6.6	<0.03
N15_4_110	Unlogged	173.0	0.1	2661.2	202.8	318.0	202.8	6.7	<0.03
N15_4_120	Unlogged	71.0	2.4	955.2	115.2	183.2	115.2	5.5	<0.03
N15_4_130	Unlogged	52.1	17.2	658.4	67.2	146.8	67.2	5.2	<0.03
N15_4_140	Unlogged	78.7	10.8	1010.8	79.6	260.8	79.6	5.4	<0.03
N15_4_150	Unlogged	68.0	2.8	900.8	90.0	199.6	90.0	5.6	1.2
N15_4_160	Unlogged	165.6	0.1	2848.0	105.6	196.4	105.6	6.4	<0.03
N15_4_170	Unlogged	212.3	0.1	3642.4	156.0	240.8	156.0	6.5	<0.03
N15_4_180	Unlogged	157.7	0.1	2541.6	188.8	216.8	188.8	6.2	<0.03
N15_4_190	Unlogged	154.0	0.1	2396.0	122.8	315.2	122.8	5.8	<0.03
N15_5_10	Unlogged	153.0	0.1	2483.2	117.6	255.2	117.6	6.0	<0.03
N15_5_20	Unlogged	201.9	0.1	3422.0	129.6	270.0	129.6	6.3	<0.03
N15_5_30	Unlogged	104.2	0.1	1572.8	110.0	219.6	110.0	6.2	<0.03
N15_5_40	Unlogged	121.8	0.1	1940.8	117.2	205.2	117.2	6.2	<0.03
N15_5_50	Unlogged	99.2	0.1	1439.6	66.8	276.8	66.8	5.9	<0.03
N15_5_60	Unlogged	134.4	0.1	2150.8	129.6	220.8	129.6	6.2	<0.03
N15_5_70	Unlogged	129.1	0.1	2120.0	90.4	208.0	90.4	6.3	<0.03
N15_5_80	Unlogged	91.6	0.1	1288.4	190.4	172.4	190.4	6.5	<0.03
N15_5_90	Unlogged	188.0	0.1	3383.6	101.6	147.2	101.6	7.1	<0.03
N15_5_100	Unlogged	311.7	0.1	5358.0	290.0	295.2	290.0	7.1	<0.03
N15_5_110	Unlogged	90.2	0.1	1179.6	164.4	242.4	164.4	5.9	<0.03
N15_5_120	Unlogged	116.5	0.1	1374.4	46.0	544.0	46.0	6.3	<0.03
N15_5_130	Unlogged	31.1	10.0	206.8	33.2	61.2	33.2	4.6	5.6
N15_5_140	Unlogged	47.0	26.8	147.6	60.8	32.4	60.8	4.3	1.2
N15_5_150	Unlogged	99.0	0.1	1439.6	134.4	217.2	134.4	5.8	<0.03
N15_5_160	Unlogged	109.9	0.1	1538.8	165.2	263.2	165.2	5.9	<0.03
N15_5_170	Unlogged	182.9	0.1	3073.6	131.2	248.8	131.2	6.3	<0.03
N15_5_180	Unlogged	195.1	0.1	3171.6	169.6	305.2	169.6	6.4	<0.03
N15_5_190	Unlogged	206.1	0.1	3618.0	150.4	184.0	150.4	6.8	<0.03
N15_5_200	Unlogged	234.1	0.1	3596.0	361.2	360.8	361.2	6.8	<0.03

For Plot ID, N3 is the logged compartment and N15 is the unlogged compartment.

Declaration

I hereby declare that this thesis was exclusively written by myself and that I have used no other sources and aids other than the ones listed.

Signature: .....

Name: JOSEPH GHARTEY

Date: 10TH OCTOBER 2024